

Exact Resonance and Coordinate Transport

Dougall–Hurwitz identities, nonlinear Stieltjes finite parts,
and polylogarithmic correlations at unequal frequencies

ProveIt research continuation

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Abstract

We answer three explicit continuation questions in the current ProveIt polylogarithm programme. A centered form of Dougall’s very-well-poised summation has only even inverse powers in its asymptotic expansion. Its Hurwitz-subtracted endpoint admits a closed formula at every nonpositive integer resonance, and every spectral derivative has a finite expression in Gamma derivatives, Hurwitz-zeta derivatives, and generalized Stieltjes constants. Reciprocal-Gamma zeros are handled before coefficient extraction. A quartic central-binomial specialization gives an all-resonance family with explicit harmonic constants; a weighted primitive reduces the hypergeometric order and transports the subtraction terms to colored polylogarithms.

A second theorem gives the exact change of a one-sided Hadamard finite part under any analytic orientation-preserving local coordinate change. The correction is a finite residue, obeys a composition law, and specializes to all argument derivatives of all generalized Stieltjes functions. For a coordinate change tangent to the identity, the correction for $\gamma_n^{(r)}$ vanishes for $n \geq 2r$, and this threshold is sharp. Finally, a finite multiplication formula reduces three unequal integer frequencies to ordinary colored Tornheim kernels, with explicit lower Stieltjes terms accounting for the fixed endpoint coordinate. It also produces convergent log-Gamma correlations and negative-order reduction identities. Every new assertion is supplied with a proof and accompanied by reproducible finite algebra or independent numerical diagnostics.

Status. The contributions are relative to the inspected repository and incoming reports. Classical summations and product-integral methods are credited explicitly. Historical priority for every derived special case is not asserted. The current Gaussian S_6 and S_8 identities remain conjectural.

Contents

1	Research scope, source baseline, and conventions	4
1.1	The questions addressed	4
1.2	Reading and audit boundaries	4
1.3	Classical antecedents	4
1.4	Conventions	5
2	Centered Dougall subtraction and every integer resonance	5
2.1	The summation family and its natural center	5
2.2	Even inverse powers and finite coefficient data	6
2.3	The coefficient differential law	7
2.4	An explicit residue polynomial and a short closed sum	8
2.5	Half-integer excess and parity cancellation	9
3	Complete Dougall jets and quartic harmonic identities	10
3.1	A holomorphic Gamma factor at the collision	10
3.2	An explicit Bell identity at zero excess	11
3.3	Finite polynomial shifts preserve every moving zero	12
3.4	Quartic central-binomial moments at all resonances	12
3.5	Generalized harmonic coefficients of every weight	13
3.6	Squared-binomial harmonic jets and log-Gamma	14
4	A hypergeometric primitive and finite polylogarithmic transport	15
4.1	A coefficientwise regularized ${}_4F_3$	15
4.2	An exact weighted primitive	16
4.3	Rational shifts give a finite colored-polylogarithm formula	16
5	Nonlinear coordinate changes of Stieltjes finite parts	17
5.1	A universal residue formula	18
5.2	Composition and the transformed singular germ	19
5.3	Every Stieltjes index and argument derivative	19
5.4	Explicit corrections through the second derivative	21
5.5	Sharp invariance and the order of the local correction	22
5.6	Polygamma functions, full derivatives, and pointwise products	23
5.7	Exact algorithm and independent checks	24
6	Three unequal frequencies with exact endpoint normalization	24
6.1	Separated singular grids and a finite lifting identity	24
6.2	A complete all-index transport formula	26
6.3	The bilinear terms are finite Stieltjes expressions	27
6.4	Ordinary colored Tornheim kernels suffice	27
7	Explicit polylogarithmic and Gamma correlation identities	29

7.1	Finite arithmetic reduction of weighted Tornheim kernels	29
7.2	An ordinary three-factor log-Gamma integral	30
7.3	A twelfth-root digamma example with explicit lower terms	31
8	Proof status, audit findings, and reproducible verification	32
8.1	What has been proved, and what is inherited	32
8.2	A verified editorial correction already proposed upstream	33
8.3	Normalization points exposed by the new proofs	33
8.4	Finite verification and its evidentiary role	34
8.5	Proposed repository integration	35
9	Further research questions and concrete next targets	35
9.1	Additional identities within the Dougall family	35
9.2	Coordinate algebra and collisions	36
9.3	Colored kernels and exact relation certificates	37
9.4	Research priorities	38
	References	38

1 Research scope, source baseline, and conventions

1.1 The questions addressed

The repository’s consolidated manuscript and its incoming research reports already contain an extensive calculus of polylogarithms, harmonic sums, Hurwitz-zeta jets, and regularized Stieltjes products. A useful continuation must retain that accumulated proof status. The baseline used here is the commit

445f754610e1377939235794de84ed9075fc09f5.

The two requested source locations are the manuscript under `Analysis/Polylogarithms/docs/manuscript` and the archives under `docs/incoming` [1].

The present work concentrates on three questions explicitly posed in the newest incoming material:

1. The Gauss–Hurwitz report asks for a fixed Dougall summation with a complete resonant coefficient law and finite Stieltjes closure [2, Section 9, Question 1].
2. The Mellin–dilation and Stieltjes–harmonic reports ask for the exact nonlinear coordinate correction to finite parts, including compatibility with composition [3, 4].
3. The Stieltjes–harmonic report asks for three unequal integer frequencies and a finite colored-kernel description with the local normalization included [4, Further research: three unequal integer frequencies].

The answer to each is constructive. The first uses even Bernoulli coefficients and a Gamma-residue identity. The second uses a local residue that depends on a finite coordinate jet. The third uses a finite Hurwitz multiplication formula before taking any finite part. In the third case, integer weights do not require a new transcendental kernel: finitely many ordinary colored Tornheim kernels suffice.

1.2 Reading and audit boundaries

The manuscript directory, its editorial ledger, relevant depth, integration, differentiation, and discovery sections, and all five incoming archives were retrieved. The complete text of the four reports most directly connected to the new identities was examined, and the fifth report’s scope and relevant source-status discussion were checked [5, 6]. The source manifest records paths and hashes; the five archive bytes were checked against their Git blob identifiers. This is a targeted proof audit, not a certification of every theorem in the 417-page source book.

The inherited results used below are rederived when the proof is short: the six-sign-region Fourier formula, the elementary Stieltjes multiplication law, and the local residue mechanism. The longer independent proof of Dougall’s summation is classical and cited [7, Eq. 16.4.9]. The existing affine correction is recovered as a specialization, which checks the normalization without presenting that earlier theorem as a fresh discovery.

1.3 Classical antecedents

Gamma-ratio asymptotics and generalized hypergeometric endpoint summation are classical. Bühring’s study of partial sums includes the zero-excess very-well-poised constant and a quartic-binomial specialization [11, Section 4]. In particular, the first quartic-binomial row below is a recovered classical identity. The extension proved here is the centered coefficient

structure, the uniform nonpositive-integer resonance law, and its complete spectral coefficient calculus with a specified Hurwitz subtraction.

Hurwitz product integrals and Tornheim sums were studied systematically by Espinosa and Moll [12, 13]; colored Tornheim relations have a separate literature [14]. The Fourier and multiplication formulas for Hurwitz zeta and the Gamma connection are standard [9]. Our unequal-frequency result is a finite transport theorem, with the fixed-coordinate terms written out. Its finite expression does not imply arithmetic independence or a minimal number of period coordinates.

Analytic regularization and Hadamard finite parts also have classical antecedents. The local theorem below is proved directly so that its Jacobian convention, residue sign, and composition rule can all be checked without an unspecified regularization prescription. The specialized Stieltjes coefficient formula and sharp high-index invariance are the concrete additions needed by the source programme.

1.4 Conventions

All powers of positive real numbers use the real logarithm. The spectral and argument derivatives are distinguished by

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(a) u^n, \quad \gamma_0(a) = -\psi(a), \quad (1.1)$$

$$\zeta^{(k)}(s, a) = \frac{\partial^k}{\partial s^k} \zeta(s, a), \quad \gamma_n^{(r)}(a) = \frac{d^r}{da^r} \gamma_n(a). \quad (1.2)$$

The same superscript on ψ denotes an argument derivative. Generalized harmonic numbers are $H_N^{(j)} = \sum_{k=1}^N k^{-j}$, with $H_0^{(j)} = 0$; $H_N = H_N^{(1)}$. The rising factorial is $(x)_N$, and $[t^k]F$ denotes a Taylor coefficient, not a derivative. Bell polynomials are normalized by

$$\exp\left(\sum_{j \geq 1} x_j \frac{t^j}{j!}\right) = \sum_{k \geq 0} Y_k(x_1, \dots, x_k) \frac{t^k}{k!}.$$

The symbol $e(x)$ means $\exp(2\pi i x)$ when introduced in the Fourier sections. On the circle, every finite part uses the right local coordinate at each singular point. The notation FP_x is reserved for that prescribed constant term; it does not denote an unnormalized analytic continuation.

The proofs justify limiting operations analytically. Exact symbolic checks verify finite identities and their implementations. Numerical checks compare independent representations and detect sign, scale, and transcription errors. They are not interval certificates or substitutes for those proofs.

2 Centered Dougall subtraction and every integer resonance

2.1 The summation family and its natural center

Fix positive real parameters a, b, c such that

$$d_0 = 1 + a - b - c > 0, \quad \eta = \frac{a}{2}, \quad x_n = n + \eta. \quad (2.1)$$

The restriction $d_0 > 0$ supplies a common domain of ordinary convergence from which all continuations below are proved. It still includes the quartic-binomial examples and all their

nonpositive-integer resonances. For $\Re u < d_0$ define

$$r_n(u) = x_n \frac{\Gamma(n+a)\Gamma(n+b)\Gamma(n+c)\Gamma(n+d_0-u)}{\Gamma(n+1)\Gamma(n+1+a-b)\Gamma(n+1+a-c)\Gamma(n+b+c+u)}. \quad (2.2)$$

All reciprocal-Gamma factors are entire functions. In particular, r_n is holomorphic when its last denominator argument passes through a nonpositive integer. No numerator Gamma pole lies in the stated half-plane.

The Rogers–Dougall summation [7, Eq. 16.4.9] gives

$$\sum_{n=0}^{\infty} r_n(u) = G(u) := \frac{\Gamma(b)\Gamma(c)\Gamma(d_0-u)\Gamma(u)}{2\Gamma(d_0)\Gamma(b+u)\Gamma(c+u)}, \quad 0 < \Re u < d_0. \quad (2.3)$$

To check the normalization, the upper parameters of the usual ${}_5F_4$ are $a, 1+a/2, b, c, d_0-u$ and its lower parameters are $a/2, 1+a-b, 1+a-c, b+c+u$. The ratio $(1+a/2)_n/(a/2)_n$ is $(n+a/2)/(a/2)$, which accounts for the factor $1/2$ in (2.3). The conventional hypergeometric excess is $2u$, whereas the Gamma factor in the sum is $\Gamma(u)$. Keeping that distinction is essential at every resonance.

2.2 Even inverse powers and finite coefficient data

Write

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \left(\frac{a}{2}, b - \frac{a}{2}, c - \frac{a}{2}, d_0 - u - \frac{a}{2} \right). \quad (2.4)$$

At the center $x_n = n + a/2$, every denominator shift is the reflection $1 - \alpha_i$ of its numerator shift:

$$r_n(u) = x_n \prod_{i=1}^4 \frac{\Gamma(x_n + \alpha_i)}{\Gamma(x_n + 1 - \alpha_i)}.$$

For Bernoulli polynomials with $B_1(z) = z - 1/2$, define

$$h_k(u) = -\frac{1}{k(2k+1)} \sum_{i=1}^4 B_{2k+1}(\alpha_i), \quad k \geq 1, \quad (2.5)$$

$$C_0(u) = 1, \quad {}^j C_j(u) = \sum_{k=1}^j k h_k(u) C_{j-k}(u), \quad j \geq 1. \quad (2.6)$$

These are polynomials with rational coefficients in a, b, c, u ; each prescribed order requires only finitely many polynomial operations.

Lemma 2.1 (Centered finite expansion). *For every $K \geq 1$,*

$$r_n(u) = x_n^{-1-2u} \left\{ \sum_{j=0}^{K-1} C_j(u) x_n^{-2j} + O(x_n^{-2K}) \right\}. \quad (2.7)$$

The remainder is locally uniform in $\Re u < d_0$ and in the positive parameter domain (2.1). Every fixed-order derivative has the corresponding differentiated expansion, with at most a fixed power of $\log x_n$.

Proof. The finite Stirling–Bernoulli expansion for the logarithm of a Gamma ratio [8, Section 5.11] contains the coefficient

$$\frac{(-1)^{k+1}}{k(k+1)} \{B_{k+1}(\alpha_i) - B_{k+1}(1 - \alpha_i)\} x_n^{-k}.$$

Bernoulli reflection makes this zero for odd k . For $k = 2\ell$ its sum over i is $h_\ell(u)$. The leading power is x_n^{-1-2u} . Exponentiation gives (2.7), and differentiation of the formal exponential in the inverse-square variable gives (2.6). This argument uses only a finite asymptotic expansion with its remainder; it does not sum a divergent asymptotic series. Uniformity follows from the uniform Stirling remainder on compact sets of shifts. To justify parameter derivatives, enlarge a given compact set slightly and apply Cauchy's formula to the remainder. The finitely many small indices are handled by the entire reciprocal-Gamma factors. $\square$

Theorem 2.2 (Centered Dougall–Hurwitz subtraction). *For $K \geq 1$, the series*

$$R_K(u) = \sum_{n=0}^{\infty} \left\{ r_n(u) - \sum_{j=0}^{K-1} C_j(u) x_n^{-1-2u-2j} \right\} \quad (2.8)$$

converges absolutely and locally normally on $-K < \Re u < d_0$ and satisfies

$$R_K(u) = G(u) - \sum_{j=0}^{K-1} C_j(u) \zeta(1 + 2u + 2j, \eta). \quad (2.9)$$

All poles of the combined right side in this strip are removable. Every fixed spectral or parameter derivative can be taken in the single bracketed series (2.8).

Proof. On a compact subset where $\Re u \geq -K + \delta$, the bracket is $O(n^{-1-2\delta})$ by Lemma 2.1. Derivatives add only fixed logarithmic powers, so the same argument gives summable majorants. The Weierstrass theorem proves holomorphy and the derivative assertion. In $0 < \Re u < d_0$, summation of the separate convergent series using (2.3) and the Hurwitz Dirichlet series gives (2.9). The identity theorem extends it to the stated strip. Its left side proves removability of the combined poles. $\square$

Remark 2.3 (The evaluated object). At $u = -N$, the two components in braces usually have divergent sums separately. Equation (2.9) evaluates their one absolutely convergent difference. The reference $\eta = a/2$ and the number of subtractions are part of the definition. Increasing that number obeys the exact relation

$$R_{K+1}(u) = R_K(u) - C_K(u) \zeta(1 + 2u + 2K, \eta), \quad -K < \Re u < d_0.$$

2.3 The coefficient differential law

The even coefficients have an exact parameter identity that considerably simplifies their resonant constant terms. Every derivative in the following operator respects $d_0 = 1 + a - b - c$:

$$V = 2\partial_a + \partial_b + \partial_c - \partial_u.$$

Lemma 2.4 (Centered coefficient transport). *For every $j \geq 1$,*

$$VC_j(u) = 2 \sum_{k=1}^j \zeta(1 - 2k, \eta) C_{j-k}(u). \quad (2.10)$$

This is a polynomial identity, since $\zeta(1 - 2k, \eta) = -B_{2k}(\eta)/(2k)$.

Proof. Temporarily regard n as a continuous positive variable. Direct logarithmic differentiation of (2.2) gives

$$Vr_n(u) = \partial_n r_n(u) + \{\psi(n+a) + \psi(n+1)\}r_n(u). \quad (2.11)$$

Here $Vx_n = \partial_n x_n = 1$. The reflected digamma pair has expansion

$$\psi(x_n + \eta) + \psi(x_n + 1 - \eta) \sim 2 \log x_n + 2 \sum_{k \geq 1} \zeta(1 - 2k, \eta) x_n^{-2k}.$$

This again follows from Bernoulli reflection in the differentiated Stirling formula. Since $Vu = -1$, the explicit exponent on the left contributes $2 \log x_n$, which cancels the displayed logarithm. The derivatives of x_n cancel between V and ∂_n . Comparison of each inverse-square coefficient proves (2.10). Uniqueness of finite asymptotic coefficients justifies this comparison. $\square$

2.4 An explicit residue polynomial and a short closed sum

For $N \geq 0$ set

$$\rho_N = \frac{(-1)^N (d_0)_N (1-b)_N (1-c)_N}{N!}. \quad (2.12)$$

Proposition 2.5 (Two finite resonance identities). *For every $N \geq 0$,*

$$C_N(-N) = \rho_N, \quad (2.13)$$

$$\frac{1}{2} C'_N(-N) + \sum_{j=0}^{N-1} C_j(-N) \zeta(1 - 2N + 2j, \eta) = \left(\partial_a + \frac{1}{2} \partial_b + \frac{1}{2} \partial_c \right) \rho_N. \quad (2.14)$$

The sum is empty when $N = 0$. The formulas hold also at all zeros of ρ_N .

Proof. Use $K = N + 1$ in Theorem 2.2. The only Hurwitz pole at $u = -N$ has index $j = N$ and residue $C_N(-N)/2$. The residue of $G(u)$ is $\rho_N/2$, using the Gamma residue and shift formulas. Since the bracketed sum is holomorphic, their residues agree. This proves (2.13) generically and then identically as a polynomial statement. Evaluate (2.10) at $u = -N$, use (2.13) for its parameter derivatives, and divide by two. This gives (2.14); the case $N = 0$ is immediate from $C_0 = \rho_0 = 1$. $\square$

Theorem 2.6 (Every nonpositive-integer resonance). *Under (2.1), for every integer $N \geq 0$,*

$$\begin{aligned} & \sum_{n=0}^{\infty} \left\{ r_n(-N) - \sum_{j=0}^N C_j(-N) x_n^{2N-1-2j} \right\} \\ &= \left[\rho_N \left[\psi(\eta) + \frac{\psi(N+1) - \psi(b) - \psi(c) - \psi(d_0 + N)}{2} \right] \right]. \end{aligned} \quad (2.15)$$

The series is absolutely convergent, including at positive integer parameters where $\rho_N = 0$.

Proof. First suppose that the shifted Gamma arguments in the following Laurent expansion are regular. With $u = -N + t$,

$$G(-N + t) = \frac{\rho_N}{2t} + \frac{\rho_N}{2} \{ \psi(N+1) - \psi(d_0 + N) - \psi(b - N) - \psi(c - N) \} + O(t).$$

The colliding Hurwitz term has constant coefficient $C'_N(-N)/2 - \rho_N \psi(\eta)$. Consequently (2.9), at $K = N + 1$, gives the claimed value except for the finite expression on the left of (2.14). The operator $W = \partial_a + (\partial_b + \partial_c)/2$ fixes d_0 , and

$$W \rho_N = \frac{\rho_N}{2} \{ \psi(b) - \psi(b - N) + \psi(c) - \psi(c - N) \}.$$

Subtraction cancels both shifted digammas and proves (2.15). Both the normally convergent sum and the final right side are locally holomorphic in the positive parameter domain. They therefore agree across zeros of ρ_N , without division by that polynomial at its zeros. $\square$

At the resonant point write $d = d_0 + N$. Then $a = b + c + d - 1 - N$ and

$$\rho_N = \frac{(1-b)_N(1-c)_N(1-d)_N}{N!},$$

which displays the symmetry of the answer in b, c, d . Our proof and theorem retain $d_0 > 0$; no continuation into an unspecified larger parameter domain is needed.

2.5 Half-integer excess and parity cancellation

Negative odd values of the conventional excess $2u$ behave differently. At $u = -M - 1/2$, the minimal centered expansion subtracts the nonnegative even powers $x_n^{2M}, x_n^{2M-2}, \dots, 1$. There is no x_n^{-1} coefficient to collide with a Hurwitz pole.

Corollary 2.7 (Negative odd hypergeometric excess). *For every integer $M \geq 0$,*

$$R_{M+1}(-M - \tfrac{1}{2}) = G(-M - \tfrac{1}{2}) - \sum_{j=0}^M C_j(-M - \tfrac{1}{2}) \zeta(-2M + 2j, \eta). \quad (2.16)$$

All terms on the right are individually regular. In particular,

$$\sum_{n=0}^{\infty} \{r_n(-\tfrac{1}{2}) - 1\} = \frac{a-1}{2} - \frac{\sqrt{\pi} \Gamma(b) \Gamma(c) \Gamma(d_0 + \tfrac{1}{2})}{\Gamma(d_0) \Gamma(b - \tfrac{1}{2}) \Gamma(c - \tfrac{1}{2})}. \quad (2.17)$$

Proof. Theorem 2.2 applies with $K = M + 1$ since $-M - 1/2 > -M - 1$. Its Hurwitz arguments are exactly the nonpositive even integers in (2.16). The case $M = 0$ uses $\Gamma(-1/2) = -2\sqrt{\pi}$ and $\zeta(0, \eta) = 1/2 - \eta$. $\square$

For example, $a = b = c = 3/4$ and $d_0 = 1/4$ give

$$\sum_{n=0}^{\infty} \left\{ \left(n + \tfrac{3}{8} \right) \left[\frac{\Gamma(n + \frac{3}{4})}{\Gamma(n + 1)} \right]^4 - 1 \right\} = -\frac{1}{8} - \sqrt{\pi} \left[\frac{\Gamma(\frac{3}{4})}{\Gamma(\frac{1}{4})} \right]^3. \quad (2.18)$$

A complementary specialization has a vanishing Gamma endpoint. For $a = b = c = 1/2$ and $u = -1/2$, put

$$b_n = \pi \frac{\binom{2n}{n}^2}{16^n}, \quad v_n = (n + \tfrac{1}{4}) b_n.$$

The two reciprocal-Gamma factors at zero give $G(-1/2) = 0$, so

$$\sum_{n \geq 0} (v_n - 1) = -\frac{1}{4}. \quad (2.19)$$

There is also a direct check: the recurrence $(n+1)^2 b_{n+1} = (n+1/2)^2 b_n$ gives $\sum_{n=0}^{M-1} v_n = M^2 b_M$. Stirling's formula yields $M^2 b_M = M - 1/4 + O(M^{-1})$, proving (2.19) independently of endpoint continuation.

At $u = -M - 1/2$, the same parameters give

$$r_n(-M - \tfrac{1}{2}) = v_n \prod_{r=0}^{M-1} \{ (n + \tfrac{1}{4})^2 - (r + \tfrac{3}{4})^2 \}, \quad G(-M - \tfrac{1}{2}) = 0.$$

Their minimally subtracted sums are rational numbers, because (2.16) involves rational polynomial coefficients and Hurwitz values at nonpositive integers and shift $1/4$. For example, with $x_n = n + 1/4$,

$$\sum_{n \geq 0} \left\{ (x_n^2 - \frac{9}{16})v_n - x_n^2 + \frac{19}{32} \right\} = \frac{21}{128}. \quad (2.20)$$

The next section develops an integer-resonance family whose values belong to $\mathbb{Q} + \pi\mathbb{Q}$.

Parity persists at an arbitrary reference shift $v > 0$, although the individual odd coefficients then need not vanish. If

$$r_n(u) \sim (n+v)^{-1-2u} \sum_{j \geq 0} A_j(u; v)(n+v)^{-j},$$

the binomial theorem gives the exact finite transport formula

$$A_j(u; v) = \sum_{\ell=0}^{\lfloor j/2 \rfloor} C_\ell(u) \frac{(1+2u+2\ell)_{j-2\ell}}{(j-2\ell)!} (v-\eta)^{j-2\ell}. \quad (2.21)$$

At the potentially colliding index it implies

$$A_{2N}(-N; v) = \rho_N, \quad A_{2N+1}(-N - \frac{1}{2}; v) = 0. \quad (2.22)$$

For the odd index every rising factorial in (2.21) contains a zero; for the even index only the term $\ell = N$ survives. Thus a generic all-powers subtraction scheme must exhibit the same residue cancellation. Derivatives of a coefficient vanishing at the resonant point need not vanish, a distinction used next.

3 Complete Dougall jets and quartic harmonic identities

3.1 A holomorphic Gamma factor at the collision

Fix $N \geq 0$. The factorization

$$E_N(t) = \frac{\Gamma(b)\Gamma(c)\Gamma(d_0 + N - t)\Gamma(1+t)}{\Gamma(d_0)\Gamma(b+t)\Gamma(c+t)} \prod_{r=1}^N \frac{(b+t-r)(c+t-r)}{t-r} \quad (3.1)$$

follows from the Gamma functional equation and satisfies

$$E_N(t) = 2tG(-N+t), \quad E_N(0) = \rho_N. \quad (3.2)$$

Every denominator in the finite product is nonzero near $t = 0$. Positive integer b or c may create numerator zeros; those zeros remain in the polynomial and are differentiated with it. Consequently (3.1) computes all Taylor coefficients without a singular Gamma quotient or a division by ρ_N .

Put

$$e_q = [t^q]E_N(t), \quad c_{j,q} = [t^q]C_j(-N+t). \quad (3.3)$$

The coefficients $c_{j,q}$ are finite polynomial data. The e_q are obtained from Bell polynomials in Gamma logarithmic derivatives at positive arguments, followed by the finite product in (3.1).

Theorem 3.1 (All resonant Stieltjes coefficients). *For $K > N$ and $m \geq 0$,*

$$[t^m]R_K(-N+t) = \frac{e_{m+1} - c_{N,m+1}}{2} - \sum_{h=0}^m c_{N,m-h} \frac{(-2)^h \gamma_h(\eta)}{h!} - \sum_{\substack{0 \leq j < K \\ j \neq N}} \sum_{q=0}^m c_{j,q} \frac{2^{m-q} \zeta^{(m-q)}(1-2N+2j, \eta)}{(m-q)!}. \quad (3.4)$$

The left side equals the absolutely convergent ordinary series

$$\sum_{n=0}^{\infty} [t^m] \left\{ r_n(-N+t) - \sum_{j=0}^{K-1} C_j(-N+t) x_n^{2N-1-2j} e^{-2t \log x_n} \right\}. \quad (3.5)$$

At $K = N+1$, the zeta derivatives in the last line of (3.4) have only the negative odd arguments $1-2N, 3-2N, \dots, -1$.

Proof. Normal convergence in Theorem 2.2 permits coefficient extraction in its bracketed series. The Gamma term is $E_N(t)/(2t)$. The singular Hurwitz factor is

$$\zeta(1+2t, \eta) = \frac{1}{2t} + \sum_{h \geq 0} \frac{(-2)^h \gamma_h(\eta)}{h!} t^h.$$

Multiplication by $C_N(-N+t)$ contributes $c_{N,m+1}/2$ and the first finite sum in (3.4). Every other Hurwitz factor is regular and expands by the ordinary Cauchy product, giving the second finite sum. The coefficient of t^{-1} cancels by (2.13). This establishes the formula without a limiting prescription at a pole. $\square$

Corollary 3.2 (Finite special-function closure). *At fixed N, m and $K = N+1$, the resonant Taylor coefficient has a finite expression in Gamma and polygamma data at $b, c, d_0 + N, 1$, generalized Stieltjes constants $\gamma_h(\eta)$ with $0 \leq h \leq m$, and spectral derivatives $\zeta^{(q)}(1-2r, \eta)$ with $1 \leq r \leq N$ and $0 \leq q \leq m$. Coefficients from finite Gamma shifts are polynomial or rational with nonsingular denominators.*

This is a representation theorem. It asserts neither algebraic independence nor a minimal special-value basis. Negative-order Hurwitz derivatives can be changed to integrated-polygamma or Barnes-type normalizations, subject to the normalization conventions already emphasized in the incoming reports [2, 4].

At a half-integer point $u_0 = -M-1/2$ there is no singular factor. Writing $\hat{c}_{j,q} = [t^q]C_j(u_0+t)$, all its jets are simply

$$[t^m]R_{M+1}(u_0+t) = [t^m]G(u_0+t) - \sum_{j=0}^M \sum_{q=0}^m \hat{c}_{j,q} \frac{2^{m-q} \zeta^{(m-q)}(-2M+2j, \eta)}{(m-q)!}. \quad (3.6)$$

Only nonpositive even spectral arguments occur; no generalized Stieltjes constant is required in this centered normalization.

3.2 An explicit Bell identity at zero excess

Define, for $k \geq 1$,

$$L_k = (-1)^k \psi^{(k-1)}(d_0) + \psi^{(k-1)}(1) - \psi^{(k-1)}(b) - \psi^{(k-1)}(c), \quad (3.7)$$

$$T_k(n) = (-1)^k \psi^{(k-1)}(n+d_0) - \psi^{(k-1)}(n+b+c). \quad (3.8)$$

The L_k are the logarithmic derivatives of $E_0(t)$, while $T_k(n)$ are those of $r_n(t)$ at $t = 0$. With the Bell normalization stated in Section 1, Theorem 3.1 gives

$$\sum_{n=0}^{\infty} \left\{ r_n(0) Y_m(T_1(n), \dots, T_m(n)) - \frac{(-2 \log x_n)^m}{x_n} \right\} = \boxed{\frac{Y_{m+1}(L_1, \dots, L_{m+1})}{2(m+1)} - (-2)^m \gamma_m(\eta)}. \quad (3.9)$$

Here $Y_0 = 1$. The formula is for derivatives; the multiplication by $m!$ relative to (3.4) explains its factor $1/(m+1)$. For instance,

$$\sum_{n \geq 0} \left\{ r_n(0) T_1(n) + \frac{2 \log x_n}{x_n} \right\} = \frac{L_1^2 + L_2}{4} + 2\gamma_1(\eta), \quad (3.10)$$

$$\sum_{n \geq 0} \left\{ r_n(0) (T_1(n)^2 + T_2(n)) - \frac{4 \log^2 x_n}{x_n} \right\} = \frac{L_1^3 + 3L_1 L_2 + L_3}{6} - 4\gamma_2(\eta). \quad (3.11)$$

The paired logarithmic counterterms are essential for convergence.

3.3 Finite polynomial shifts preserve every moving zero

There is a convenient exact identity at the summand level:

$$\boxed{r_n(-N+t) = r_n(t)(n+d_0-t)_N(n+b+c-N+t)_N}. \quad (3.12)$$

It follows from the Gamma functional equation. The base $r_n(t)$ has positive Gamma arguments at $t = 0$; the finite polynomial contains every shifted reciprocal-Gamma zero. Thus all resonant summand jets are finite convolutions of its polynomial coefficients with the Bell coefficients in (3.8). This provides a direct implementation for all orders without singular harmonic expressions at small n .

The distinction is substantive. A term that is zero after substituting $u = -N$ can have nonzero first and higher derivatives. Deleting such terms before constructing the deformation changes the value of the jet. The following family supplies a simple exact example.

3.4 Quartic central-binomial moments at all resonances

Set $a = b = c = d_0 = 1/2$ and put

$$x_n = n + \frac{1}{4}, \quad w_n = x_n \left[\frac{\Gamma(n + \frac{1}{2})}{\Gamma(n+1)} \right]^4 = \pi^2 x_n \frac{\binom{2n}{n}^4}{256^n}, \quad (3.13)$$

$$P_N(X) = \prod_{r=0}^{N-1} \{X - (r + \frac{1}{4})^2\}. \quad (3.14)$$

The finite Gamma shift gives

$$r_n(-N) = w_n P_N(x_n^2), \quad \rho_N = \frac{(-1)^N (\frac{1}{2})_N^3}{N!}. \quad (3.15)$$

Using

$$\psi(\frac{1}{4}) = -\gamma - 3 \log 2 - \frac{\pi}{2}, \quad \psi(N + \frac{1}{2}) = -\gamma - 2 \log 2 + 2H_{2N} - H_N,$$

Theorem 2.6 becomes the all-resonance identity

$$\sum_{n=0}^{\infty} \left\{ w_n P_N(x_n^2) - \sum_{j=0}^N C_j(-N) x_n^{2N-1-2j} \right\} = \frac{(-1)^N \left(\frac{1}{2}\right)_N^3}{N!} \left(H_N - H_{2N} - \frac{\pi}{2} \right). \quad (3.16)$$

All counterterms still follow from the finite recurrence (2.5)–(2.6); for example,

$$C_1(u) = \frac{u^3}{3} + \frac{u^2}{4} - \frac{u}{48} - \frac{1}{16}. \quad (3.17)$$

They can also be built from one universal base sequence. If σ_k denotes the elementary symmetric polynomial in $(1/4)^2, (5/4)^2, \dots, (N-3/4)^2$, with $\sigma_0 = 1$, then

$$C_j(-N) = \sum_{k=0}^{\min(j,N)} (-1)^k \sigma_k C_{j-k}(0). \quad (3.18)$$

This follows by multiplying the polynomial $P_N(x_n^2)$ into the base expansion of w_n . Its first base coefficients are $C_0(0) = 1$, $C_1(0) = -1/16$, $C_2(0) = 3/256$, and $C_3(0) = -23/4096$. Thus successive moment subtractions do not require an independent asymptotic calculation. The first three cases are

$$\sum_{n \geq 0} \left\{ w_n - \frac{1}{x_n} \right\} = -\frac{\pi}{2}, \quad (3.19)$$

$$\sum_{n \geq 0} \left\{ n(n + \tfrac{1}{2}) w_n - x_n + \frac{1}{8x_n} \right\} = \frac{\pi + 1}{16}, \quad (3.20)$$

$$\sum_{n \geq 0} \left\{ n(n-1)(n + \tfrac{1}{2})(n + \tfrac{3}{2}) w_n - x_n^3 + \frac{27}{16} x_n - \frac{27}{128 x_n} \right\} = -\frac{63 + 54\pi}{512}. \quad (3.21)$$

The zero-excess case (3.19) is a recovered classical Ramanujan–Dougall constant, explicitly included in Bühring’s partial-sum analysis [11, Section 4, Corollaries 1–2]. The uniform resonant mechanism and its complete jets are the continuation supplied here. The displayed higher examples are proved specializations, without a claim that each has no earlier appearance.

3.5 Generalized harmonic coefficients of every weight

For the same half-parameter family, write

$$g = \gamma + \log 2, \quad g_3 = \gamma + 3 \log 2.$$

Then $T_1(n) = 2(g - H_{2n})$, $L_1 = 2g_3$, and for $k \geq 2$,

$$T_k(n) = (k-1)! \{ (2^k - 1 - (-1)^k) \zeta(k) - 2^k H_{2n}^{(k)} + (1 + (-1)^k) H_n^{(k)} \}, \quad (3.22)$$

$$L_k = (k-1)! \{ (1 - 2(-1)^k)(2^k - 1) + (-1)^k \} \zeta(k). \quad (3.23)$$

These follow by substituting the standard half-integer polygamma values into (3.7)–(3.8). In particular, $L_2 = -2\zeta(2)$, $L_3 = 40\zeta(3)$, and $L_4 = -84\zeta(4)$. Together with (3.9), they give an explicit all-weight family of generalized harmonic identities. Its first nonconstant member is

$$\sum_{n \geq 0} \left\{ w_n (g - H_{2n}) + \frac{\log x_n}{x_n} \right\} = \frac{g_3^2}{2} - \frac{\zeta(2)}{4} + \gamma_1\left(\frac{1}{4}\right). \quad (3.24)$$

At higher resonance, the stronger finite identity

$$r_n(-N+t) = r_n(t) \prod_{r=0}^{N-1} \{x_n^2 - (r + \frac{1}{4} - t)^2\} \quad (3.25)$$

supplies all harmonic summand coefficients by finite convolution. For $n < N$, put $k = N - n - 1$. The first revived term is

$$[t]r_n(-N+t) = w_n(n + \frac{1}{2})_N n! (-1)^k k!. \quad (3.26)$$

Indeed, the simple zero of $1/\Gamma(-k+t)$ has leading coefficient $(-1)^k k! t$; the other factors are regular. In particular,

$$r_0(-1) = 0, \quad [t]r_0(-1+t) = \frac{\pi^2}{8}. \quad (3.27)$$

For completeness, the full next-resonance first derivative can be written without any exceptional small index. For every $n \geq 0$,

$$[t]r_n(-1+t) = w_n \{2n(n + \frac{1}{2})(g - H_{2n}) + \frac{1}{2}\}.$$

Also

$$C_1(-1+t) = -\frac{1}{8} + \frac{23}{48}t - \frac{3}{4}t^2 + \frac{1}{3}t^3.$$

Substituting these coefficients in (3.4) yields

$$\begin{aligned} & \sum_{n \geq 0} \left\{ w_n [2n(n + \frac{1}{2})(g - H_{2n}) + \frac{1}{2}] + 2x_n \log x_n - \frac{\frac{1}{4} \log x_n + \frac{23}{48}}{x_n} \right\} \\ &= -\frac{g_3^2}{8} + \frac{7g_3}{48} + \frac{\zeta(2)}{16} - \frac{1}{16} - \frac{23\pi}{96} - \frac{\gamma_1(\frac{1}{4})}{4} - 2\zeta^{(1)}(-1, \frac{1}{4}). \end{aligned} \quad (3.28)$$

One way to verify the constant directly is to use $E_1(t) = E_0(t)(t - 1/2)^3/(1-t)$ in (3.1), and

$$[t^2]E_1(t) = -\frac{L_1^2}{16} + \frac{5L_1}{8} - \frac{L_2}{16} - \frac{7}{8}.$$

The formula combines a generalized harmonic summand, a Stieltjes constant, a negative-odd Hurwitz derivative, and the revived finite head. Deleting the $n = 0$ summand after first substituting $u = -1$ would change the first derivative by exactly $\pi^2/8$.

3.6 Squared-binomial harmonic jets and log-Gamma

The squared-binomial half-resonance family in (2.19) has an additional cancellation. At $a = b = c = 1/2$,

$$G(-\frac{1}{2}+t) = \frac{\sqrt{\pi} \Gamma(1-t) \Gamma(-\frac{1}{2}+t)}{2 \Gamma(t)^2} = -\pi t^2 + O(t^3). \quad (3.29)$$

Thus its constant and first Taylor coefficients vanish. Recall $v_n = \pi(n + 1/4) \binom{2n}{n}^2 / 16^n$, $x_n = n + 1/4$, and $g = \gamma + \log 2$. The logarithmic derivatives of $r_n(-1/2+t)$ at $t = 0$ are

$$2(g - H_{2n}), \quad -2\zeta(2) + 4H_{2n}^{(2)} - 2H_n^{(2)}$$

at orders one and two. Applying (3.6) with $M = 0$, and using Lerch's identity for $\zeta^{(1)}(0, 1/4)$ [9, Eq. 25.11.18], gives

$$\boxed{\sum_{n \geq 0} \{v_n(g - H_{2n}) + \log x_n\} = \frac{1}{2} \log(2\pi) - \log \Gamma(\frac{1}{4})}. \quad (3.30)$$

The second derivative gives

$$\sum_{n \geq 0} \left\{ v_n [(g - H_{2n})^2 - \tfrac{1}{2}\zeta(2) + H_{2n}^{(2)} - \tfrac{1}{2}H_n^{(2)}] - \log^2 x_n \right\} = -\frac{\pi}{2} - \zeta^{(2)}(0, \tfrac{1}{4}). \quad (3.31)$$

Both are single absolutely convergent bracketed sums. Equations (3.6) and (3.29) similarly determine every higher harmonic jet. Here the pole-free half-resonance produces Hurwitz derivatives at zero, while the integer-resonance family (3.9) produces generalized Stieltjes constants at the pole. The distinction follows from the exact centered spectrum.

4 A hypergeometric primitive and finite polylogarithmic transport

4.1 A coefficientwise regularized ${}_4F_3$

The centered prefactor has a useful consequence for antiderivatives. For the same parameter domain, set

$$q_n(u) = \frac{r_n(u)}{n + \eta}, \quad Q_u(z) = \sum_{n=0}^{\infty} q_n(u) z^n, \quad |z| < 1. \quad (4.1)$$

Generically this is the Gamma-normalized hypergeometric function

$$Q_u(z) = \frac{\Gamma(a)\Gamma(b)\Gamma(c)\Gamma(d_0 - u)}{\Gamma(1 + a - b)\Gamma(1 + a - c)\Gamma(b + c + u)} \times {}_4F_3 \left(\begin{matrix} a, b, c, d_0 - u \\ 1 + a - b, 1 + a - c, b + c + u \end{matrix}; z \right). \quad (4.2)$$

At a nonpositive lower parameter, the product in (4.2) is defined by the coefficient series (4.1), or equivalently by regularizing the lower Gamma factors before evaluation. The ordinary ${}_4F_3$ and its prefactor must not be separately evaluated as an undefined expression times zero. The definition (4.1) retains the zero summands and all their parameter derivatives.

Let $\theta = z \, d/dz$ and use the Lerch transcendent [10, Section 25.14]

$$\Phi(z, s, \eta) = \sum_{n=0}^{\infty} \frac{z^n}{(n + \eta)^s}, \quad |z| < 1.$$

The subtracted generating function is

$$F_{K,u}(z) = (\theta + \eta)Q_u(z) - \sum_{j=0}^{K-1} C_j(u)\Phi(z, 1 + 2u + 2j, \eta). \quad (4.3)$$

It is exactly the generating function of the brackets in (2.8). For $-K < \Re u < d_0$, their absolute summability gives

$$\lim_{z \uparrow 1} F_{K,u}(z) = R_K(u). \quad (4.4)$$

The limit and every fixed spectral derivative are locally uniform on compact subsets of this strip. The identity follows from the summable majorant in Theorem 2.2, not merely from an unqualified appeal to boundary continuation.

4.2 An exact weighted primitive

Theorem 4.1 (The ${}_5F_4$ -to- ${}_4F_3$ primitive). *For $|z| < 1$ on a fixed branch of z^η , the normalized radial primitive of (4.3) is*

$$\int_0^z t^{\eta-1} F_{K,u}(t) dt = z^\eta \left\{ Q_u(z) - \sum_{j=0}^{K-1} C_j(u) \Phi(z, 2 + 2u + 2j, \eta) \right\}. \quad (4.5)$$

It vanishes at zero. Every fixed spectral derivative and every parameter derivative within the domain may be taken through this identity, including the dependence of the weight on $\eta = a/2$.

Proof. The defining series converge normally for $|z| < 1$ on compact parameter sets. Integrating the coefficient of z^n uses

$$\int_0^z t^{n+\eta-1} dt = \frac{z^{n+\eta}}{n+\eta}.$$

Thus the coefficient $r_n(u) = (n+\eta)q_n(u)$ loses its centered factor, while the order of every Lerch coefficient increases by one. This proves (4.5). The condition $\eta > 0$ makes the integrand locally integrable at zero and the displayed primitive zero there. On slightly enlarged compact parameter sets, additional logarithmic factors from differentiating $t^{\eta-1}$ are still integrable. Normal convergence and dominated integration therefore give the derivative assertions. $\square$

More generally, division of coefficients by $(n+\eta)^\ell$ gives successive applications of the normalized inverse of $\theta + \eta$:

$$F_{K,u}^{(\ell)}(z) = \sum_{n=0}^{\infty} \frac{r_n(u) z^n}{(n+\eta)^\ell} - \sum_{j=0}^{K-1} C_j(u) \Phi(z, 1 + 2u + 2j + \ell, \eta), \quad \ell \geq 0, \quad (4.6)$$

$$(\theta + \eta) F_{K,u}^{(\ell+1)} = F_{K,u}^{(\ell)}. \quad (4.7)$$

Here $F_{K,u}^{(0)} = F_{K,u}$ and the first step is the lower-order hypergeometric expression in (4.5). All these identities are coefficientwise regular at moving lower Gamma zeros.

4.3 Rational shifts give a finite colored-polylogarithm formula

Let $q \geq 1$, $1 \leq p \leq q$, and $\eta = p/q$. Choose a local q th root $w = z^{1/q}$ and put $\omega = e^{2\pi i/q}$. Root-of-unity filtering gives

$$z^{p/q} \Phi(z, s, p/q) = q^{s-1} \sum_{\ell=0}^{q-1} \omega^{-p\ell} \text{Li}_s(\omega^\ell w). \quad (4.8)$$

To prove it, expand $\text{Li}_s(\omega^\ell w)$ for $|w| < 1$ and use

$$\frac{1}{q} \sum_{\ell=0}^{q-1} \omega^{\ell(k-p)} = \begin{cases} 1, & k \equiv p \pmod{q}, \\ 0, & \text{otherwise.} \end{cases}$$

The surviving positive indices are $k = qn + p$, including $p = q$ with $n \geq 0$, which yields (4.8). The formula is entire in the spectral order s for $|z| < 1$.

For $m \geq 0$, write $\partial_s^m \text{Li}_s$ for order derivatives, not argument derivatives. Differentiation of (4.8) gives

$$z^{p/q} \partial_s^m \Phi(z, s, p/q) = q^{s-1} \sum_{\ell=0}^{q-1} \omega^{-p\ell} \sum_{h=0}^m \binom{m}{h} (\log q)^{m-h} \partial_s^h \text{Li}_s(\omega^\ell w). \quad (4.9)$$

Thus every rational-shift subtraction term and every spectral jet in (4.5) is a finite combination of classical polylogarithms and their order derivatives. At integer resonances the primitive orders are the even integers $2 - 2N + 2j$; at the half-integer points they are the odd integers $1 - 2M + 2j$. Nonpositive integer polylogarithms themselves reduce to rational functions by repeated application of $z d/dz$ to $z/(1-z)$, while their order derivatives retain additional information.

For a larger rational shift $\eta = L + p/q$, $L \geq 0$ integer, one must first apply the finite-head identity

$$z^\eta \Phi(z, s, \eta) = z^{p/q} \Phi(z, s, p/q) - \sum_{r=0}^{L-1} \frac{z^{r+p/q}}{(r+p/q)^s}. \quad (4.10)$$

This follows by removing the first L terms and is valid for every s . Differentiating it retains the corresponding finite logarithmic terms. Dropping this finite head would invalidate the claimed polylogarithmic reduction for $\eta > 1$.

For the quartic family $\eta = 1/4$, the four colors are $1, i, -1, -i$:

$$z^{1/4} \Phi(z, s, 1/4) = 4^{s-1} \{ \text{Li}_s(w) - i \text{Li}_s(iw) - \text{Li}_s(-w) + i \text{Li}_s(-iw) \}, \quad w = z^{1/4}. \quad (4.11)$$

Together with (3.25) and (4.5), this gives a finite Gaussian-colored polylogarithm description of the counterterms for every resonant quartic moment and all its harmonic derivatives. It does not furnish the missing exact relation certificates for the manuscript's conjectural S_6 or S_8 evaluations.

Remark 4.2 (Boundary values of the primitive). In the strip of Theorem 2.2, the bracketed coefficients in (4.3) are absolutely summable. Dividing them by $n + \eta$ preserves this property. Therefore the primitive itself has a finite radial endpoint, obtainable from its one combined series. At a resonant parameter, however, some separate terms on the right of (4.5) can be singular at $z = 1$. Their combined expression, or its normally convergent coefficient series, is the appropriate endpoint object. No separate assignment of values to divergent polylogarithmic pieces is implicit.

5 Nonlinear coordinate changes of Stieltjes finite parts

The incoming Mellin–dilation and Stieltjes–harmonic reports explicitly ask for the replacement of a local affine coordinate qt by $f(t) = qt + c_2 t^2 + \dots$, including every Stieltjes index, every argument derivative, and a composition law [3, 4]. The answer is a finite residue formula. Besides recovering their affine polynomial, it yields a sharp additional phenomenon: if $f'(0) = 1$, the finite part of $\gamma_n^{(r)}$ transforms without a local correction whenever $n \geq 2r$.

The method below is direct analytic regularization. Its general finite-part principle is not presented as a claim of historical priority. The contribution to the stated source question is the complete formula, its specialization to the Stieltjes hierarchy, the exact composition law, and the sharp filtration of the surviving local terms.

5.1 A universal residue formula

All coordinates in this section are local real coordinates, with the singular approach from the right. Let

$$h(x) = \sum_{\ell=0}^L a_\ell(x)(\log x)^\ell, \quad x > 0, \quad (5.1)$$

where the a_ℓ are meromorphic germs at zero, each with a finite pole order. Define $\text{FP}_x h$ by the constant term of $\int_\varepsilon h(x)\phi(x) dx$ as $\varepsilon \downarrow 0$, for a smooth compactly supported test function ϕ . Every negative power of ε and every nonconstant power of $\log \varepsilon$ is discarded. The positive-side germ can be extended by zero on the left; a fixed smooth completion can equally well be added. Such a completion changes neither the residue nor the coordinate correction.

Let f be a real-analytic local diffeomorphism with

$$f(0) = 0, \quad f'(0) = q > 0, \quad \lambda_f(t) = \log \frac{f(t)}{t}. \quad (5.2)$$

The last function is analytic at zero, with $\lambda_f(0) = \log q$. Pullback here means scalar distributional pullback:

$$\langle f^*T, \phi \rangle = \langle T, (\phi \circ f^{-1})(f^{-1})' \rangle. \quad (5.3)$$

In particular, the pullback of an ordinary function is its composition with f .

Theorem 5.1 (Nonlinear finite-part coordinate law). *For the germ (5.1), define*

$$\langle \mathfrak{A}_f[h], \phi \rangle = \sum_{\ell=0}^L \frac{1}{\ell+1} \text{Res}_{t=0} a_\ell(f(t)) \lambda_f(t)^{\ell+1} \phi(t) dt. \quad (5.4)$$

Then

$$\boxed{f^* \text{FP}_x h = \text{FP}_t(h \circ f) + \mathfrak{A}_f[h].} \quad (5.5)$$

If every a_ℓ has pole order at most R , the correction is a linear combination of $\delta_0, \delta'_0, \dots, \delta_0^{(R-1)}$; for $R = 0$ it is zero. Every coefficient is determined by finite Taylor data. The residue notation for a smooth ϕ means its finite Taylor coefficient; holomorphic extension of the test function is unnecessary.

Proof. It suffices to take $h(x) = a(x) \log^\ell x$. For $\Re z$ sufficiently large, the pairing of $x_+^z a(x)$ with a test function is an ordinary integral. If $a(x) = x^{-R} A(x)$ with A holomorphic, subtract the Taylor polynomial of $A(x)\phi(x)$ through degree $R-1$. The remainder is locally integrable for z near zero, uniformly on a small closed disk. This constructs a meromorphic distribution with at most a simple pole at zero, and

$$\text{FP}_x(a(x) \log^\ell x) = \ell! [z^\ell] (x_+^z a(x)). \quad (5.6)$$

Here the bracket is a Laurent coefficient. To check the normalization, the only exceptional Taylor monomial is x^{z-1} , whose continued integral to a fixed positive h_0 is h_0^z/z . Its coefficient $[z^\ell]$ is $\log^{\ell+1} h_0/(\ell+1)!$, precisely the constant remaining after subtracting the lower-endpoint logarithm. Every other Taylor monomial has a denominator nonzero at $z = 0$.

After pullback the regulated integrand is

$$t^z a(f(t)) e^{z\lambda_f(t)} \phi(t).$$

Its polar Taylor term at zero has numerator

$$B(z) = \text{Res}_{t=0} a(f(t)) e^{z\lambda_f(t)} \phi(t) dt. \quad (5.7)$$

Thus the exceptional contribution is $B(z)h_0^z/z$. On the other hand, the cutoff integral of the same term is $B(z)(h_0^z - \varepsilon^z)/z$. In its coefficient $[z^\ell]$, the lower endpoint has constant term $[z^{\ell+1}]B(z)$: all its other terms are positive powers of $\log \varepsilon$. Consequently the meromorphic coefficient exceeds the coefficientwise cutoff constant by

$$\ell! [z^{\ell+1}]B(z) = \frac{1}{\ell+1} \operatorname{Res}_{t=0} a(f(t))\lambda_f(t)^{\ell+1}\phi(t) dt.$$

Nonpolar Taylor terms have lower-endpoint powers different from zero and introduce no constant discrepancy. The subtracted remainder has uniform integrable majorants, so Cauchy's coefficient formula justifies the coefficient extraction there. This proves (5.5).

Finally, $a_\ell(f(t))$ has pole order at most R and λ_f is analytic. Only derivatives $\phi^{(j)}(0)$ with $j \leq R-1$ can occur in its residue. This proves both the support and finite-data claims. $\square$

5.2 Composition and the transformed singular germ

Theorem 5.2 (Exact residue cocycle). *For two orientation-preserving analytic coordinate germs f, g ,*

$$\mathfrak{A}_{g \circ f}[h] = f^*\mathfrak{A}_g[h] + \mathfrak{A}_f[h \circ g]. \quad (5.8)$$

The last term acts on the transformed germ $h \circ g$, which again has the form (5.1). This dependence is part of the composition law.

Proof. The assertion follows by applying Theorem 5.1 twice. There is also an explicit coefficient proof. For a single term $h(x) = a(x) \log^\ell x$, put $A = \lambda_f(t)$ and $B = \lambda_g(f(t))$. Then

$$\lambda_{g \circ f} = A + B, \quad h(g(y)) = \sum_{k=0}^{\ell} \binom{\ell}{k} a(g(y)) \lambda_g(y)^{\ell-k} (\log y)^k.$$

Residue substitution $y = f(t)$ in the pullback (5.3) shows that $f^*\mathfrak{A}_g[h]$ contributes the factor $B^{\ell+1}/(\ell+1)$. The other term contributes

$$\sum_{k=0}^{\ell} \frac{\binom{\ell}{k}}{k+1} B^{\ell-k} A^{k+1} = \frac{(A+B)^{\ell+1} - B^{\ell+1}}{\ell+1}.$$

The sum is precisely (5.4) for $g \circ f$. This finite polynomial identity proves composition with all scale and nonlinear terms retained. $\square$

For later implementation, the same residue convention gives

$$\langle f^*\delta_0^{(r)}, \phi \rangle = (-1)^r r! \operatorname{Res}_{t=0} f(t)^{-r-1} \phi(t) dt. \quad (5.9)$$

Indeed, substitution $x = f(t)$ changes the residue into the coefficient of x^r in $(\phi \circ f^{-1})(f^{-1})'$, in agreement with the definition of $\delta_0^{(r)}$. Equations (5.8) and (5.9) provide an exact composition algorithm. A formula that retained h unchanged in the last term of (5.8) would generally be incorrect.

5.3 Every Stieltjes index and argument derivative

The local singularity is fixed by the Hurwitz convention:

$$\gamma_n(x) = \frac{\log^n x}{x} + \gamma_n(1+x), \quad \gamma_0(x) = -\psi(x). \quad (5.10)$$

The second term is analytic near zero. Set

$$T_{n,r} = \text{FP}_x \gamma_n^{(r)}(x), \quad G_n = T_{n,0}. \quad (5.11)$$

Locally these may be regarded as the periodic finite parts used in [3, 4]; the smooth completion in (5.10) is then the same on both sides of the endpoint. Every comparison involving $D^r G_n$ below uses this common smooth periodic completion. In particular, the analytic remainder is smooth across zero and contributes no step-function derivatives.

For $r \geq 0$, let

$$\mathcal{E}_r(z) = \prod_{j=1}^r (1 - z/j), \quad \mathcal{E}_0(z) = 1, \quad (5.12)$$

$$\begin{aligned} b_{n,r}(L) &= n![z^{n+1}]e^{Lz}\mathcal{E}_r(z), \\ \alpha_{n,r}(L) &= b_{n,r}(L) - b_{n,r}(0) = n![z^{n+1}](e^{Lz} - 1)\mathcal{E}_r(z). \end{aligned} \quad (5.13)$$

If $e_j(r)$ is the elementary symmetric polynomial of degree j in $1, 1/2, \dots, 1/r$, with $e_0(r) = 1$ and $e_j(r) = 0$ for $j > r$, then

$$\alpha_{n,r}(L) = n! \sum_{j=0}^{\min(r,n)} \frac{(-1)^j e_j(r)}{(n+1-j)!} L^{n+1-j}. \quad (5.14)$$

The subtraction of $b_{n,r}(0)$ is necessary because $T_{n,r}$ in (5.11) is the finite part of a pointwise derivative.

Theorem 5.3 (Nonlinear Stieltjes transport). *For every $n, r \geq 0$,*

$$f^* T_{n,r} = \text{FP}_t \gamma_n^{(r)}(f(t)) + \mathfrak{A}_{n,r}(f), \quad (5.15)$$

where

$$\langle \mathfrak{A}_{n,r}(f), \phi \rangle = (-1)^r r! \text{Res}_{t=0} f(t)^{-r-1} \alpha_{n,r}(\lambda_f(t)) \phi(t) dt. \quad (5.16)$$

In particular,

$$\mathfrak{A}_{n,r}(f) = \sum_{j=0}^r c_{n,r,j}(f) \delta_0^{(j)}, \quad (5.17)$$

$$c_{n,r,j}(f) = \frac{(-1)^{r+j} r!}{j!} [t^{r-j}] \left(\frac{f(t)}{t} \right)^{-r-1} \alpha_{n,r} \left(\log \frac{f(t)}{t} \right). \quad (5.18)$$

The coordinate jet through degree $r+1$ suffices for every Stieltjes index n .

Proof. Differentiate x^{z-1} in its argument:

$$\partial_x^r x^{z-1} = (-1)^r r! \mathcal{E}_r(z) x^{z-r-1}.$$

Taking $n![z^n]$ gives the exact singular polynomial

$$\partial_x^r \frac{\log^n x}{x} = (-1)^r r! x^{-r-1} \sum_{j=0}^{\min(r,n)} \frac{(-1)^j e_j(r) n!}{(n-j)!} \log^{n-j} x. \quad (5.19)$$

Apply Theorem 5.1 term by term. Its factor $1/(n-j+1)$ converts the displayed polynomial into (5.14), while the analytic remainder in (5.10) contributes no residue. Expanding the test function through degree r yields (5.18), since $\langle \delta_0^{(j)}, \phi \rangle = (-1)^j \phi^{(j)}(0)$. The coefficient in that formula uses the expansion of $f(t)/t$ only through degree r , proving the last assertion. $\square$

For comparison, distributional differentiation gives

$$D^r G_n = T_{n,r} + b_{n,r}(0)\delta_0^{(r)}, \quad b_{n,r}(0) = (-1)^{n+1} n! e_{n+1}(r). \quad (5.20)$$

This also follows by the regulator computation: at identity scale the polar numerator for $\partial_x^r x^{z-1}$ is $\mathcal{E}_r(z)\delta_0^{(r)}$. Thus

$$f^* D^r G_n = \text{FP}_t \gamma_n^{(r)}(f(t)) + \mathfrak{C}_{n,r}(f), \quad (5.21)$$

$$\langle \mathfrak{C}_{n,r}(f), \phi \rangle = (-1)^r r! \text{Res}_{t=0} f(t)^{-r-1} b_{n,r}(\lambda_f(t)) \phi(t) dt. \quad (5.22)$$

The finite coefficient formula for $\mathfrak{C}$ is (5.18) with α replaced by b .

Corollary 5.4 (Recovery of the affine contact polynomial). *At $f(t) = qt$, $q > 0$,*

$$\mathfrak{A}_{n,r}(qt) = q^{-r-1} \alpha_{n,r}(\log q) \delta_0^{(r)}, \quad (5.23)$$

$$D_t^r f^* G_n = \text{FP}_t (q^r \gamma_n^{(r)}(qt)) + \frac{b_{n,r}(\log q)}{q} \delta_0^{(r)}. \quad (5.24)$$

The second identity is exactly the local form of the incoming integer-dilation theorem, including its normalization by $1/q$.

Proof. The function $f(t)/t$ is constant. Only the $j = r$ term survives in (5.18); also $D_t^r f^* G_n = q^r f^* D^r G_n$. $\square$

5.4 Explicit corrections through the second derivative

Write

$$f(t) = qt(1 + at + bt^2 + O(t^3)), \quad L = \log q. \quad (5.25)$$

Thus $a = f''(0)/(2q)$ and $b = f'''(0)/(6q)$. The first coefficient polynomials are

$$\begin{aligned} \alpha_{0,r}(L) &= L, \\ \alpha_{1,r}(L) &= \frac{L^2}{2} - H_r L, \\ \alpha_{2,r}(L) &= \frac{L^3}{3} - H_r L^2 + (H_r^2 - H_r^{(2)})L. \end{aligned} \quad (5.26)$$

Here $H_0 = H_0^{(2)} = 0$. At $r = 0$, the entire answer is

$$\mathfrak{A}_{n,0}(f) = \frac{L^{n+1}}{q(n+1)} \delta_0. \quad (5.27)$$

Nonlinear coefficients first affect argument derivatives.

For $r = 1$ set $\alpha = \alpha_{n,1}(L)$; primes below differentiate this polynomial in L . Equation (5.18) becomes

$$\mathfrak{A}_{n,1}(f) = q^{-2} [\alpha \delta_0' + a(2\alpha - \alpha') \delta_0]. \quad (5.28)$$

For $r = 2$, with $\alpha = \alpha_{n,2}(L)$, it gives

$$\begin{aligned} \mathfrak{A}_{n,2}(f) &= q^{-3} [\alpha \delta_0'' + 2a(3\alpha - \alpha') \delta_0' \\ &\quad + \{(12a^2 - 6b)\alpha + (2b - 7a^2)\alpha' + a^2 \alpha''\} \delta_0]. \end{aligned} \quad (5.29)$$

To verify these formulas directly, use

$$\lambda_f = L + at + (b - a^2/2)t^2 + O(t^3), \quad (1 + at + bt^2)^{-3} = 1 - 3at + (6a^2 - 3b)t^2 + O(t^3).$$

The same displays with $b_{n,r}$ in place of $\alpha_{n,r}$ give the correction $\mathfrak{C}_{n,r}$ for distributional derivatives.

At unit tangent, these are particularly short:

$$\mathfrak{A}_{n,1} = \begin{cases} -a\delta_0, & n = 0, \\ a\delta_0, & n = 1, \\ 0, & n \geq 2, \end{cases} \quad (5.30)$$

and

$$\mathfrak{A}_{n,2} = \begin{cases} -2a\delta'_0 + (2b - 7a^2)\delta_0, & n = 0, \\ 3a\delta'_0 + (-3b + \frac{23}{2}a^2)\delta_0, & n = 1, \\ -2a\delta'_0 + (2b - 10a^2)\delta_0, & n = 2, \\ 3a^2\delta_0, & n = 3, \\ 0, & n \geq 4. \end{cases} \quad (5.31)$$

These expressions retain curvature and the cubic coordinate coefficient even though the linear scale has disappeared.

Example 5.5 (A direct cutoff check of the curvature terms). Take $f(t) = t + at^2 + bt^3 + O(t^4)$ and $\gamma''_0(x)$, whose singular part is $2x^{-3}$. Its inverse is $g(x) = x - ax^2 + (2a^2 - b)x^3 + O(x^4)$. For a test function equal to one near zero, pullback uses the inverse Jacobian $g'(x) = 1 - 2ax + (6a^2 - 3b)x^2 + O(x^3)$. A primitive of the terms that can contribute a finite constant is

$$-x^{-2} + 4ax^{-1} + (12a^2 - 6b)\log x.$$

It has zero constant in the x coordinate. After $x = f(\varepsilon)$ its constant is $(2b - 3a^2) - 4a^2 = 2b - 7a^2$, giving the δ_0 coefficient in (5.31). For a test function equal to t , the inverse multiplier is $g(x)g'(x) = x - 3ax^2 + O(x^3)$; the relevant primitive is $-2/x - 6a\log x$. Its composed constant is $2a$, which equals the action of $-2a\delta'_0$ on that test function. This calculation uses cutoff primitives directly and confirms both nonlinear signs without invoking the residue formula.

5.5 Sharp invariance and the order of the local correction

Theorem 5.6 (Sharp Stieltjes-index filtration). *Suppose $f'(0) = 1$, put $a = f''(0)/2$, and let $r \geq 1$. Then $\mathfrak{A}_{n,r}(f)$ is a linear combination of $\delta_0^{(j)}$ with*

$$0 \leq j \leq \min\{r - 1, 2r - n - 1\}. \quad (5.32)$$

An empty range means the zero distribution. In particular,

$$\boxed{f^* \text{FP}_x \gamma_n^{(r)} = \text{FP}_t \gamma_n^{(r)}(f(t)) \quad (n \geq 2r).} \quad (5.33)$$

For $r \leq n \leq 2r - 1$, let $d = n + 1 - r$ and $k = 2r - n - 1$. The coefficient of the highest possible derivative is

$$[\delta_0^{(k)}]\mathfrak{A}_{n,r}(f) = \frac{(-1)^k n!}{d! k!} a^d. \quad (5.34)$$

For $0 \leq n \leq r$, the coefficient of $\delta_0^{(r-1)}$ is

$$[\delta_0^{(r-1)}]\mathfrak{A}_{n,r}(f) = (-1)^{n+1} r n! e_n(r) a. \quad (5.35)$$

The two formulas agree at $n = r$. If $a \neq 0$, every stated upper order is attained. At the last nonzero index,

$$\mathfrak{A}_{2r-1,r}(f) = \frac{(2r-1)!}{r!} a^r \delta_0. \quad (5.36)$$

For $r = 0$, every correction vanishes at unit tangent by (5.27).

Proof. Now $\lambda_f(t) = at + O(t^2)$. Formula (5.14) shows that the smallest possible power of L in $\alpha_{n,r}$ is $\max(1, n+1-r)$. Therefore the analytic factor multiplying t^{-r-1} in (5.16) vanishes to at least that order. Its residue against ϕ can use only derivatives of orders at most $r - \max(1, n+1-r)$, proving (5.32) and (5.33).

For $r \leq n \leq 2r-1$ the lowest term is, using $e_r(r) = 1/r!$,

$$\alpha_{n,r}(L) = \frac{(-1)^r n!}{r! d!} L^d + O(L^{d+1}).$$

To produce $\delta_0^{(k)}$ in (5.18), only the coefficient of $t^{r-k} = t^d$ is required. It is the displayed coefficient times a^d ; the constant term of $(f(t)/t)^{-r-1}$ is one. Substitution proves (5.34). For $n \leq r$ the linear coefficient of $\alpha_{n,r}$ is $(-1)^n n! e_n(r)$. Taking $j = r-1$ in the same formula proves (5.35). All these elementary symmetric coefficients are nonzero in their indicated ranges. Equation (5.36) is the case $d = r, k = 0$. $\square$

Remark 5.7 (Higher tangency gives stronger invariance). If $f(t) = t + O(t^{d+1})$, then $\lambda_f = O(t^d)$. If $d > r$, every $\mathfrak{A}_{n,r}$ vanishes. If $1 \leq d \leq r$, it suffices that

$$n \geq r + \lfloor r/d \rfloor.$$

This threshold is sharp in general. For $f(t) = t + ct^{d+1} + O(t^{d+2})$, set $h = \lfloor r/d \rfloor$, $n = r + h - 1$ and $k = r - dh$. The coefficient of $\delta_0^{(k)}$ is $(-1)^k n! c^h / (h! k!)$. This follows from the same lowest-power argument.

5.6 Polygamma functions, full derivatives, and pointwise products

For polygamma, the correction in (5.15) is $-\mathfrak{A}_{0,r}$ because $\gamma_0^{(r)} = -\psi^{(r)}$. For example, at $f(t) = t + at^2 + O(t^3)$,

$$f^* \text{FP}_x \psi'(x) = \text{FP}_t \psi'(f(t)) + a\delta_0. \quad (5.37)$$

The undifferentiated function $\log \Gamma(x)$ is locally integrable; its logarithmic singularity has no negative-power coefficient in (5.1). Its ordinary pullback therefore has no anomaly.

A derivative in the argument of γ_n differs from a full derivative of $\gamma_n(f(t))$. The latter is also finite and explicit. Let $B_{k,r}$ denote the partial exponential Bell polynomial in the ordinary Faà di Bruno formula. For $k \geq 1$,

$$\begin{aligned} D_t^k f^* G_n - \text{FP}_t D_t^k (\gamma_n(f(t))) \\ = \sum_{r=1}^k B_{k,r}(f', f'', \dots, f^{(k-r+1)}) \mathfrak{C}_{n,r}(f). \end{aligned} \quad (5.38)$$

Indeed, the distributional chain rule holds first for smooth functions and then by pullback and differentiation of distributions; it gives the displayed Bell sum with $f^* D^r G_n$. Finite parts commute with multiplication by a smooth function, directly from their test-function definition, so (5.21) proves (5.38). To put the answer in constant delta coordinates, use the finite identity

$$v\delta_0^{(j)} = \sum_{\ell=0}^j (-1)^{j-\ell} \binom{j}{\ell} v^{(j-\ell)}(0) \delta_0^{(\ell)}. \quad (5.39)$$

Thus no indefinite distribution operation remains in the formula.

The universal theorem also treats a pointwise product once its own finite part has been specified. This gives a small exact illustration relevant to collision normalizations.

Corollary 5.8 (Coordinate law for a coincident Stieltjes product). *Let $h(x) = \gamma_m(x)\gamma_n(x)$, $M = m + n$, and use (5.25). Then*

$$\begin{aligned} \mathfrak{A}_f[h] = & -\frac{L^{M+1}}{q^2(M+1)}\delta'_0 \\ & + \left[\frac{a}{q^2} \left(L^M - \frac{2L^{M+1}}{M+1} \right) + \frac{\gamma_n(1)L^{m+1}}{q(m+1)} + \frac{\gamma_m(1)L^{n+1}}{q(n+1)} \right] \delta_0. \end{aligned} \quad (5.40)$$

As usual $L^0 = 1$, also when $L = 0$. At unit tangent the sole nonzero case is $m = n = 0$, with correction $a\delta_0$.

Proof. Equation (5.10) gives the complete nonintegrable part

$$\frac{\log^M x}{x^2} + \frac{\gamma_n(1+x)\log^m x + \gamma_m(1+x)\log^n x}{x}.$$

For the first term, expand $f^{-2}\lambda_f^{M+1}/(M+1)$ through the coefficient of t^{-1} . It gives the first line of (5.40) and its a/q^2 term. The two simple-pole terms require only the values $\gamma_n(1)$ and $\gamma_m(1)$, and yield the remaining terms. The analytic product remainder has no residue. $\square$

This corollary compares coordinate choices for a specified finite part of an ordinary pointwise product. It does not identify that finite part with a collision limit of separated products, and it does not define multiplication of arbitrary distributions at common singular supports. It supplies explicit local normalization data for such a future comparison.

5.7 Exact algorithm and independent checks

For fixed n, r , the complete computation consists of four finite operations: form $\mathcal{E}_r$, extract $\alpha_{n,r}$, truncate $\log(f(t)/t)$ and $(f(t)/t)^{-r-1}$ through degree r , and read the coefficients in (5.18). The coefficients are rational polynomials in the normalized coordinate derivatives and $\log q$, with the overall factor q^{-r-1} . Composition is checked by the binomial identity in Theorem 5.2.

The accompanying program `code/verify_nonlinear.py` performs exact rational symbolic checks of the coefficient generator, the scale-aware cocycle, the generic first- and second-derivative formulas, and the sharp filtration. It also compares the correction with an independent direct cutoff-primitive calculation for finite test jets. The record is `results/nonlinear_checks.json`. These finite tests check the implementation and normalization; the all-index analytic assertions are established by the proofs above.

6 Three unequal frequencies with exact endpoint normalization

The incoming three-factor formula treats translated copies of the same periodic Hurwitz function. Its next question allows three distinct positive integer frequencies [4]. Solving the Fourier constraint directly produces weighted linear forms. A finite multiplication identity gives a shorter route: first replace each integer-frequency factor by finitely many unit-frequency translates. All local scales can then be accounted for before the product is formed.

6.1 Separated singular grids and a finite lifting identity

Fix $p_1, p_2, p_3 \in \mathbb{N}_{>0}$ and $0 \leq a_j < 1$. Put

$$\mathcal{B}_j = \left\{ b_{j,h} = \frac{a_j + h}{p_j} : 0 \leq h < p_j \right\} \subset [0, 1). \quad (6.1)$$

Assume the three sets $\mathcal{B}_j$ are pairwise disjoint. The singular points of the j th original factor are $x = -b$ modulo one, with $b \in \mathcal{B}_j$; hence this assumption is precisely disjointness of the singular grids. It is directly decidable for rational data. In a pairwise arithmetic form, if $d = \gcd(p_i, p_j)$, it says

$$(p_i/d)a_j - (p_j/d)a_i \notin \mathbb{Z} \quad (i \neq j).$$

The grid formulation will make the local geometry transparent.

Lemma 6.1 (Finite periodic multiplication). *For a positive integer p , $0 \leq a < 1$, and an argument away from the singular grid,*

$$\zeta(s, \{px + a\}) = p^{-s} \sum_{h=0}^{p-1} \zeta\left(s, \left\{x + \frac{a+h}{p}\right\}\right). \quad (6.2)$$

Both sides agree as meromorphic functions of s . If $L = \log p$, then for every $m \geq 0$,

$$\gamma_m(\{px + a\}) = \frac{1}{p} \sum_{h=0}^{p-1} \sum_{\ell=0}^m \binom{m}{\ell} L^{m-\ell} \gamma_\ell\left(\left\{x + \frac{a+h}{p}\right\}\right) - \frac{L^{m+1}}{m+1}. \quad (6.3)$$

Proof. Write $y = \{px + a\}$. The p numbers $\{x + (a+h)/p\}$, counted with multiplicity, are a permutation of $(y+k)/p$, $0 \leq k < p$. Splitting the defining Hurwitz series into residue classes gives $\sum_k \zeta(s, (y+k)/p) = p^s \zeta(s, y)$ for $\Re s > 1$. Meromorphic continuation proves (6.2).

Set $s = 1 - z$. The expansion $\zeta(1 - z, y) = -1/z + \sum_{m \geq 0} \gamma_m(y) z^m / m!$ shows that the nonpolar part on the right of (6.2) is

$$\frac{e^{Lz}}{p} \sum_{h=0}^{p-1} \sum_{\ell \geq 0} \gamma_\ell(\{x + (a+h)/p\}) \frac{z^\ell}{\ell!} - \frac{e^{Lz} - 1}{z}.$$

Its m th derivative at zero is (6.3). This derivation also fixes the sign and the absence of a factor $1/p$ in the last term. $\square$

Let G_m be the unit-coordinate periodic finite part of γ_m . It has mean zero. For completeness,

$$\gamma_m(x) = x^{-1} \log^m x + \gamma_m(1+x)$$

has a singular term of finite-part integral zero on $(0, 1)$. Also $\int_1^2 \zeta(1 - z, x) dx = -1/z$, by $\partial_x \zeta(-z, x) = z \zeta(1 - z, x)$ and the Hurwitz shift relation. Its regular coefficients give $\int_1^2 \gamma_m(x) dx = 0$. Therefore $\langle G_m, 1 \rangle = 0$.

Taking the constant term of the pointwise identity (6.3) in the same ambient coordinate at every singular point gives an equality of distributions. Define

$$\begin{aligned} c_m(p) &= \frac{\log^{m+1} p}{m+1}, \\ W_{p,a;m} &= \frac{1}{p} \sum_{h=0}^{p-1} \sum_{\ell=0}^m \binom{m}{\ell} (\log p)^{m-\ell} G_\ell(x + (a+h)/p). \end{aligned} \quad (6.4)$$

Then

$$\text{FP}_x \gamma_m(\{px + a\}) = W_{p,a;m} - c_m(p). \quad (6.5)$$

In particular,

$$\text{FP}_x \int_0^1 \gamma_m(\{px + a\}) dx = -\frac{\log^{m+1} p}{m+1}. \quad (6.6)$$

This nonzero mean is a simple check on the endpoint scale. It would be missed by replacing a fixed-coordinate finite part by an uncorrected pullback of a mean-zero periodic distribution.

6.2 A complete all-index transport formula

For pairwise distinct unit-frequency shifts b_1, b_2, b_3 , let

$$I_\ell(\mathbf{b}) = \text{FP}_x \int_0^1 \prod_{j=1}^3 \gamma_{\ell_j}(\{x + b_j\}) dx. \quad (6.7)$$

For $0 < c < 1$, write

$$J_{uv}(c) = \text{FP}_x \int_0^1 \gamma_u(x) \gamma_v(\{x + c\}) dx. \quad (6.8)$$

All finite parts use the right coordinate $x - x_0$ at each singularity. Different cutoffs may tend independently to zero. Since the singular supports are disjoint, their distribution products are defined by ordinary multiplication by smooth functions near each singularity; their integrals equal these finite parts.

Fix $m_1, m_2, m_3 \geq 0$, and abbreviate

$$L_j = \log p_j, \quad c_j = \frac{L_j^{m_j+1}}{m_j + 1}, \quad A_{j,\ell} = \frac{1}{p_j} \binom{m_j}{\ell} L_j^{m_j-\ell}.$$

As usual, a zeroth power is one, also when $L_j = 0$.

Theorem 6.2 (All-index unequal-frequency reduction). *Under the disjoint-grid hypothesis, the ambient-coordinate finite part*

$$I_m^p(\mathbf{a}) = \text{FP}_x \int_0^1 \prod_{j=1}^3 \gamma_{m_j}(\{p_j x + a_j\}) dx$$

has the finite expression

$$\begin{aligned} I_m^p(\mathbf{a}) = & \sum_{h_1=0}^{p_1-1} \sum_{h_2=0}^{p_2-1} \sum_{h_3=0}^{p_3-1} \sum_{\ell_1=0}^{m_1} \sum_{\ell_2=0}^{m_2} \sum_{\ell_3=0}^{m_3} \left(\prod_{j=1}^3 A_{j,\ell_j} \right) I_\ell(b_{1,h_1}, b_{2,h_2}, b_{3,h_3}) \\ & - \sum_{k=1}^3 c_k \sum_{h_i=0}^{p_i-1} \sum_{h_j=0}^{p_j-1} \sum_{u=0}^{m_i} \sum_{v=0}^{m_j} A_{i,u} A_{j,v} J_{uv}(\{b_{j,h_j} - b_{i,h_i}\}) - c_1 c_2 c_3. \end{aligned} \quad (6.9)$$

In the k th term of the second line, $i < j$ are the other two indices. Every pair argument lies in $(0, 1)$, and every triple has distinct arguments. Thus no collision continuation is hidden in this formula.

Proof. Apply (6.5) to each of the three factors. Since different grids have disjoint singular support, their product can be expanded distributively, with no undefined distribution product. The term containing three W factors gives the first line of (6.9). The three terms containing two W factors give its second line. The terms containing one W factor integrate to zero because each G_ℓ has zero mean. The constant term is $-c_1 c_2 c_3$. This proves the formula. Equivalently, a partition of unity at the separated endpoints proves the same identity directly by linearity of the cutoff constant terms. $\square$

This theorem resolves the stated unequal-frequency question at every Stieltjes index. It permits a redundant finite list of coordinates rather than claiming a minimal period basis. Integer frequencies are essential to the finite multiplication step. Colliding grids require a separately specified pointwise product finite part or joint regularization.

6.3 The bilinear terms are finite Stieltjes expressions

The pair terms in (6.9) are already reduced in the incoming reports [4, 3]. We record their complete coefficient formula so that the triple transport is constructive without an unspecified lower-order operation. Put

$$\begin{aligned} \mathcal{G}(z; c) &= \zeta(1 - z, c) + 1/z, & U(z, w) &= \frac{\Gamma(1 + z)\Gamma(1 - z - w)}{\Gamma(1 - w)}, \\ V(z, w) &= \frac{U(z, w) - 1}{z(z + w)}, & r &= z + w. \end{aligned} \quad (6.10)$$

Both $\mathcal{G}$ and V are holomorphic at the indicated origin. For V , the numerator vanishes on $z = 0$ and on $z = -w$; these are distinct linear factors. The formula is

$$\begin{aligned} \sum_{u, v \geq 0} J_{uv}(c) \frac{z^u w^v}{u!v!} &= \frac{\mathcal{G}(r; c) - \mathcal{G}(w; c)}{z} + \frac{\mathcal{G}(r; 1 - c) - \mathcal{G}(z; 1 - c)}{w} \\ &\quad - V(z, w) - V(w, z) \\ &\quad + r\{V(z, w)\mathcal{G}(r; c) + V(w, z)\mathcal{G}(r; 1 - c)\}. \end{aligned} \quad (6.11)$$

Every apparent singularity is removable. In particular,

$$J_{00}(c) = \gamma_1(c) + \gamma_1(1 - c) - 2\zeta(2). \quad (6.12)$$

Here is a short derivation. Fourier expansion and Gamma reflection give the meromorphic identity

$$\begin{aligned} \int_0^1 \zeta(1 - z, x)\zeta(1 - w, \{x + c\}) dx &= B(z, 1 - z - w)\zeta(1 - z - w, c) \\ &\quad + B(w, 1 - z - w)\zeta(1 - z - w, 1 - c). \end{aligned}$$

It first holds in an absolutely convergent Fourier domain and extends to $\Re z, \Re w > 0$ by endpoint integrability. Its Laurent decomposition at zero is

$$-\frac{1}{zw} + \frac{\mathcal{G}(w; c)}{z} + \frac{\mathcal{G}(z; 1 - c)}{w} + \sum_{u, v \geq 0} J_{uv}(c) \frac{z^u w^v}{u!v!}.$$

One obtains this decomposition by subtracting the singular Taylor term at each of the two disjoint endpoints. Substitute $B(z, 1 - r) = U(z, w)/z$ and $U = 1 + zrV$ in the beta expression and subtract the three displayed polar terms. The remainder is exactly (6.11). Finally,

$$U(z, w) = \exp\left(\sum_{k \geq 2} \frac{\zeta(k)}{k} \{(-1)^k z^k + (z + w)^k - w^k\}\right)$$

shows that any requested pair coefficient uses only finitely many Stieltjes constants and polynomial combinations of positive integer zeta values. No infinite coefficient search is required.

6.4 Ordinary colored Tornheim kernels suffice

For $|X| = |Y| = 1$, $X, Y \neq 1$, define initially in an absolutely convergent region

$$\mathcal{T}(A, B, C; X, Y) = \sum_{m, n \geq 1} \frac{X^m Y^n}{m^A n^B (m + n)^C}. \quad (6.13)$$

Its derivatives in A, B, C are called spectral derivatives. They are not derivatives of the fixed colors.

Lemma 6.3 (Entire continuation and exact negative-order values). *The kernel (6.13) has a joint entire continuation in (A, B, C) . For $\Re C > 0$ it is given by*

$$\mathcal{T}(A, B, C; X, Y) = \frac{1}{\Gamma(C)} \int_0^\infty t^{C-1} \text{Li}_A(Xe^{-t}) \text{Li}_B(Ye^{-t}) dt. \quad (6.14)$$

For every $N \geq 0$,

$$\mathcal{T}(A, B, -N; X, Y) = \sum_{j=0}^N \binom{N}{j} \text{Li}_{A-j}(X) \text{Li}_{B-N+j}(Y). \quad (6.15)$$

Proof. The Gamma integral for $(m+n)^{-C}$ proves (6.14) in a common absolutely convergent domain. Since $X, Y \neq 1$, each polylogarithm is analytic in t in a neighborhood of zero and entire in its order. Their product has a Taylor expansion $f(t) = \sum_{j=0}^M f_j t^j + O_K(t^{M+1})$, uniformly on compact order sets. Split the integral at one, subtract this polynomial on $(0, 1)$, and add $\sum_{j=0}^M f_j/(C+j)$. The resulting expression extends to $\Re C > -M-1$ after multiplication by $1/\Gamma(C)$. The simple Gamma zeros cancel the apparent poles $C = 0, -1, \dots, -M$. The integral at infinity is locally normal because of exponential decay. Increasing M proves joint entire continuation. At $C = -N$ the value is $(-1)^N f^{(N)}(0)$; applying $\partial_t \text{Li}_A(Xe^{-t}) = -\text{Li}_{A-1}(Xe^{-t})$ and Leibniz's rule gives (6.15). $\square$

For pairwise distinct b_1, b_2, b_3 , let $i < j$ be the indices other than k and put

$$X_k = e(b_i - b_k), \quad Y_k = e(b_j - b_k), \quad \theta_k = \frac{\pi}{2}(\lambda_i + \lambda_j - \lambda_k), \quad e(t) = e^{2\pi i t}.$$

Define the finite sum

$$\begin{aligned} S(\boldsymbol{\lambda}; \mathbf{b}) &= \sum_{k=1}^3 \{e^{-i\theta_k} \mathcal{T}(\lambda_i, \lambda_j, \lambda_k; X_k, Y_k) \\ &\quad + e^{i\theta_k} \mathcal{T}(\lambda_i, \lambda_j, \lambda_k; X_k^{-1}, Y_k^{-1})\}. \end{aligned} \quad (6.16)$$

The signs specify all six regions of the three-frequency hyperplane. The identity rederived next is the unit-frequency result already established in the incoming Stieltjes-harmonic report.

Proposition 6.4 (The unit-frequency spectral identity). *For $\Re \lambda_j > 0$,*

$$K_1(\boldsymbol{\lambda}; \mathbf{b}) := \int_0^1 \prod_{j=1}^3 \zeta(1 - \lambda_j, \{x + b_j\}) dx = \frac{\prod_j \Gamma(\lambda_j)}{(2\pi)^{\sum_j \lambda_j}} S(\boldsymbol{\lambda}; \mathbf{b}). \quad (6.17)$$

Consequently

$$I_\ell(\mathbf{b}) = \left(\prod_{j=1}^3 \ell_j! \right) [\lambda_1^{\ell_1+1} \lambda_2^{\ell_2+1} \lambda_3^{\ell_3+1}] \left\{ \prod_{j=1}^3 \frac{\Gamma(1 + \lambda_j)}{(2\pi)^{\lambda_j}} S(\boldsymbol{\lambda}; \mathbf{b}) \right\}. \quad (6.18)$$

Proof. For $\Re \lambda_j > 1$, the Fourier coefficients of each factor are

$$\widehat{Z}_{1-\lambda}(n) = \frac{\Gamma(\lambda)}{(2\pi|n|)^\lambda} e^{-i\pi\lambda \text{sgn}(n)/2}, \quad n \neq 0, \quad \widehat{Z}_{1-\lambda}(0) = 0.$$

This is Hurwitz's classical Fourier formula [9, Eq. 25.11.9]. Absolute convergence permits termwise integration. The surviving triples satisfy $n_1 + n_2 + n_3 = 0$, with every $n_j \neq 0$. Either the unique negative index is k , giving $n_i = m, n_j = n, n_k = -m - n$, or all signs are reversed. Their denominators and phases are precisely (6.16). Near each separated singular point, only

one factor behaves like t^{λ_j-1} ; the others are smooth. Local integrable majorants extend the identity to $\Re \lambda_j > 0$.

To extract the finite parts, split at disjoint neighborhoods of the three singularities. The singular factor is $t^{\lambda_k-1} + \zeta(1 - \lambda_k, 1 + t)$. For a smooth multiplier g , the continued integral is

$$\frac{g(0)h^{\lambda_k}}{\lambda_k} + \int_0^h t^{\lambda_k-1}(g(t) - g(0)) dt.$$

The second term is holomorphic near zero after clearing the at most simple poles in the other order variables. The Laurent coefficient $[\lambda_k^{\ell_k}]$ of the first term is the unit-coordinate finite part divided by $\ell_k!$. Taylor subtraction gives uniform remainders $O(\varepsilon^{1-\rho})$ on a closed polydisk $|\lambda_j| \leq \rho < 1$ after multiplication by $\lambda_1 \lambda_2 \lambda_3$. Cauchy's formula therefore justifies simultaneous coefficient extraction. Finally, $\lambda_j \Gamma(\lambda_j) = \Gamma(1 + \lambda_j)$ converts (6.17) to (6.18). $\square$

Combining Theorem 6.2, (6.11), and (6.18) is the promised finite formula in standard colored kernels. Its largest triple spectral derivative order is at most $m_1 + m_2 + m_3 + 3$.

Corollary 6.5 (Spectral transport before taking finite parts). *For $\Re \lambda_j > 0$,*

$$\begin{aligned} K_p(\boldsymbol{\lambda}; \mathbf{a}) &:= \int_0^1 \prod_{j=1}^3 \zeta(1 - \lambda_j, \{p_j x + a_j\}) dx \\ &= \left(\prod_{j=1}^3 p_j^{\lambda_j-1} \right) \sum_{h_1=0}^{p_1-1} \sum_{h_2=0}^{p_2-1} \sum_{h_3=0}^{p_3-1} K_1(\boldsymbol{\lambda}; b_{1,h_1}, b_{2,h_2}, b_{3,h_3}). \end{aligned} \quad (6.19)$$

Both sides have the same meromorphic continuation. The spectral coefficient at the Stieltjes pole must still be converted to the named ambient-coordinate finite part; (6.9) performs that conversion explicitly.

Proof. Multiply three instances of (6.2) with $s = 1 - \lambda_j$, then integrate. The sums are finite and every resulting triple is separated. Proposition 6.4 provides the continuation. $\square$

7 Explicit polylogarithmic and Gamma correlation identities

The preceding transport theorem is an all-index result. This section makes several consequences explicit, including an ordinary convergent Gamma integral and a root-of-unity example whose lower terms reduce to classical Stieltjes constants and logarithms.

7.1 Finite arithmetic reduction of weighted Tornheim kernels

The weighted double kernel that appears in a direct Fourier treatment is itself a finite transform of (6.13). Let $p, q \geq 1$ be integers and define

$$\mathcal{T}_{p,q}(A, B, C; X, Y) = \sum_{m,n \geq 1} \frac{X^m Y^n}{m^A n^B (pm + qn)^C} \quad (7.1)$$

in an absolutely convergent region. Fix any roots $\xi^p = X$ and $\eta^q = Y$.

Proposition 7.1 (A finite root filter). *If $|X| = |Y| = 1$ and $X, Y \neq 1$, then*

$$\mathcal{T}_{p,q}(A, B, C; X, Y) = p^{A-1} q^{B-1} \sum_{h=0}^{p-1} \sum_{k=0}^{q-1} \mathcal{T}(A, B, C; \xi e(h/p), \eta e(k/q)). \quad (7.2)$$

The expression is independent of the chosen roots and is entire in (A, B, C) . In particular, for $N \geq 0$,

$$\mathcal{T}_{p,q}(A, B, -N; X, Y) = \sum_{j=0}^N \binom{N}{j} p^j q^{N-j} \text{Li}_{A-j}(X) \text{Li}_{B-N+j}(Y). \quad (7.3)$$

All A - and B -derivatives of this last identity are valid.

Proof. Substitute $M = pm$ and $N' = qn$ in (7.1). The indicator of $p \mid M$ is $p^{-1} \sum_h e(hM/p)$, and similarly for $q \mid N'$. The factors $m^{-A} = p^A M^{-A}$ and $n^{-B} = q^B (N')^{-B}$ give (7.2). Changing a chosen root permutes the finite sum. No new color equals one because $X, Y \neq 1$; hence Lemma 6.3 proves entire continuation. Expanding $(pm + qn)^N$ in the original convergent region proves (7.3), and continuation gives the full claim. $\square$

Finite congruence conditions on the two indices can be imposed by a further finite root filter. This explains why integral weights and congruence restrictions arising from the unequal-frequency lattice can be handled without introducing a new family of transcendental kernels. It does not assert that the finite list produced in this way has no further relations.

The derivative in the *third* order requires separate data. Differentiating an identity known only at $C = -N$ is not legitimate. An exact formula specifying that additional data follows from the same Mellin continuation. Put $f(t) = \text{Li}_A(Xe^{-t}) \text{Li}_B(Ye^{-t})$ and $f_j = [t^j]f(t)$. Then

$$\begin{aligned} \partial_C \mathcal{T}(A, B, C; X, Y)|_{C=-N} = (-1)^N N! \left\{ \int_0^1 t^{-N-1} \left(f(t) - \sum_{j=0}^N f_j t^j \right) dt + \int_1^\infty t^{-N-1} f(t) dt \right. \\ \left. + \sum_{j=0}^{N-1} \frac{f_j}{j-N} - \psi(N+1) f_N \right\}. \end{aligned} \quad (7.4)$$

Both integrals converge ordinarily. To prove this, use the Taylor subtraction in Lemma 6.3 and

$$\frac{1}{\Gamma(-N+v)} = (-1)^N N! v \{1 - \psi(N+1)v + O(v^2)\}.$$

The pole f_N/v supplies the last term in (7.4). The formula displays exactly what remains after the negative-order value reduces to products of polylogarithms. It is an integral representation for the transverse derivative, not a claim that this derivative reduces to the value at $C = -N$.

7.2 An ordinary three-factor log-Gamma integral

Define the centered log-Gamma function and a constant by

$$\Lambda(x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi) = \zeta'(0, x), \quad \kappa = \gamma + \log(2\pi). \quad (7.5)$$

Its periodic version is integrable to every finite power near an endpoint. The identity below therefore requires no finite part. For a variable A and an integer $p \geq 1$, use the operators

$$D_{+,A}^{(p)} = \partial_A + \log p - \kappa - \frac{i\pi}{2}, \quad D_{-,A}^{(p)} = \partial_A + \log p - \kappa + \frac{i\pi}{2}. \quad (7.6)$$

Theorem 7.2 (Unequal-frequency centered Gamma correlation). *For the disjoint grids (6.1),*

$$\begin{aligned} \int_0^1 \prod_{j=1}^3 \Lambda(\{p_j x + a_j\}) dx \\ = -\frac{1}{4\pi^3} \Im \sum_{h_1=0}^{p_1-1} \sum_{h_2=0}^{p_2-1} \sum_{h_3=0}^{p_3-1} \sum_{k=1}^3 D_{+,A}^{(p_i)} D_{+,B}^{(p_j)} D_{-,C}^{(p_k)} \mathcal{T}(A, B, C; X_k, Y_k) \Big|_{A=B=C=1}. \end{aligned} \quad (7.7)$$

Here $i < j$ are the other indices, and X_k, Y_k are formed from the lifted triple $(b_{1,h_1}, b_{2,h_2}, b_{3,h_3})$ as in (6.16). The imaginary part is taken only after specialization to these real spectral orders.

Proof. Lerch's Gamma identity in (7.5) gives

$$\int_0^1 \prod_j \Lambda(\{p_j x + a_j\}) dx = -\partial_{\lambda_1} \partial_{\lambda_2} \partial_{\lambda_3} K_p(\boldsymbol{\lambda}; \mathbf{a})|_{\boldsymbol{\lambda}=(1,1,1)}.$$

Differentiation under the integral is justified, near each separated endpoint, by a majorant of the form $t^{-\rho}(1 + |\log t|^3)$ with a fixed $0 < \rho < 1$ in a sufficiently small neighborhood of that spectral point. Insert (6.19) and (6.17). At $\lambda = 1$,

$$p^{\lambda-1} \Gamma(\lambda) (2\pi)^{-\lambda} = (2\pi)^{-1}, \quad \partial_\lambda \log\{p^{\lambda-1} \Gamma(\lambda) (2\pi)^{-\lambda}\} = \log p - \kappa.$$

The phase in a region with two positive frequencies is $-i$ at this point. Its derivatives contribute $-i\pi/2$ in the two positive positions and $+i\pi/2$ in the negative one, giving the operators (7.6). The opposite region is the complex conjugate after evaluating at real orders. Thus its pair contributes $2\Re(-iW) = 2\Im W$. The common factor $(2\pi)^{-3}$ and the initial minus sign give (7.7). $\square$

The formula is a finite Gamma-polylogarithm identity with ordinary integrals and entire spectral derivatives. The correlation of three uncentered $\log \Gamma$ factors follows by expanding $\log \Gamma = \Lambda + \frac{1}{2} \log(2\pi)$. Since $\int_0^1 \Lambda(\{px + a\}) dx = 0$, the only additional terms are the three ordinary centered bilinear correlations and $\frac{1}{8} \log^3(2\pi)$. Those bilinear correlations already have single-polylogarithm order-derivative evaluations in the incoming dilation report [4].

7.3 A twelfth-root digamma example with explicit lower terms

For pairwise distinct b_1, b_2, b_3 modulo one, let

$$\mathcal{P}(b_1, b_2, b_3) = \text{FP}_x \int_0^1 \prod_{j=1}^3 \psi(\{x + b_j\}) dx.$$

The unit-frequency formula (6.18) gives

$$\mathcal{P}(\mathbf{b}) = -2\Re \sum_{k=1}^3 D_{+,A}^{(1)} D_{+,B}^{(1)} D_{-,C}^{(1)} \mathcal{T}(A, B, C; X_k, Y_k) \Big|_{A=B=C=0}. \quad (7.8)$$

This is the already established translated three-digamma coordinate formula; our example uses it after unequal-frequency transport.

Choose

$$(p_1, p_2, p_3) = (1, 2, 3), \quad (a_1, a_2, a_3) = (0, \frac{1}{3}, \frac{1}{4}). \quad (7.9)$$

The lifted sets are

$$\mathcal{B}_1 = \{0\}, \quad \mathcal{B}_2 = \{\frac{1}{6}, \frac{2}{3}\}, \quad \mathcal{B}_3 = \{\frac{1}{12}, \frac{5}{12}, \frac{3}{4}\}.$$

They are disjoint, and every color in (7.8) is a twelfth root of unity distinct from one. Theorem 6.2, with $m_1 = m_2 = m_3 = 0$ and $\gamma_0 = -\psi$, gives the exact identity

$$\begin{aligned} \text{FP}_x \int_0^1 \psi(x) \psi(\{2x + \frac{1}{3}\}) \psi(\{3x + \frac{1}{4}\}) dx \\ = \frac{1}{6} \sum_{b \in \mathcal{B}_2} \sum_{c \in \mathcal{B}_3} \mathcal{P}(0, b, c) + \mathcal{L}, \end{aligned} \quad (7.10)$$

$$\mathcal{L} := \frac{\log 2}{3} \sum_{c \in \mathcal{B}_3} J_{00}(c) + \frac{\log 3}{2} \sum_{b \in \mathcal{B}_2} J_{00}(b). \quad (7.11)$$

The last constant reduces completely. With $u = \log 2$, $v = \log 3$, and $\gamma_1 = \gamma_1(1)$,

$$\begin{aligned} \mathcal{L} = & 2(\gamma_1 - \zeta(2))(u + v) - \gamma(6u^2 + 4uv + 3v^2) \\ & - 7u^3 - 7u^2v - 4uv^2 - \frac{3}{2}v^3. \end{aligned} \quad (7.12)$$

To verify the reduction, Stieltjes multiplication gives

$$\begin{aligned} \gamma_0\left(\frac{1}{4}\right) + \gamma_0\left(\frac{3}{4}\right) &= 2\gamma + 6\log 2, \\ \gamma_1\left(\frac{1}{4}\right) + \gamma_1\left(\frac{3}{4}\right) &= 2\gamma_1 - 6\gamma\log 2 - 7\log^2 2, \\ \gamma_0\left(\frac{1}{3}\right) + \gamma_0\left(\frac{2}{3}\right) &= 2\gamma + 3\log 3, \\ \gamma_1\left(\frac{1}{3}\right) + \gamma_1\left(\frac{2}{3}\right) &= 2\gamma_1 - 3\gamma\log 3 - \frac{3}{2}\log^2 3. \end{aligned}$$

Apply the same multiplication law to the two lifted grids in (7.11) and insert (6.12). Collecting terms gives (7.12).

Independent subtracted quadrature gives the following diagnostic values:

$$\text{FP}_x \int_0^1 \psi(x)\psi(\{2x + \tfrac{1}{3}\})\psi(\{3x + \tfrac{1}{4}\}) \, dx \approx 77.3418540125851504884052200711,$$

$$\mathcal{L} \approx -23.0290680355012727975296691002.$$

These are numerical illustrations of the proved identities. Their normalization is the stated ambient finite part. In particular, removing the lower terms in (7.11) would change the value by the displayed nonzero constant.

8 Proof status, audit findings, and reproducible verification

8.1 What has been proved, and what is inherited

The distinction between a proved identity, a finite representation, and a numerical candidate is essential to this continuation. The following table records the status of its principal components.

Result	Status here	Precise scope
Dougall summation and the zero-excess binomial constant	Classical input	Dougall supplies (2.3); the constant (3.19) is already in the Ramanujan–Bühning literature.
Centered coefficients and every integer resonance	Proved continuation	Theorems 2.2, 2.6, and 3.1 hold in the stated positive parameter domain, including residue zeros.
Harmonic moment identities and primitives	Proved specializations	All displayed sums are single convergent brackets. The primitive is coefficientwise regular at moving lower Gamma zeros.
Nonlinear Stieltjes coordinate change	Proved continuation	Theorems 5.1, 5.2, 5.3, and 5.6 supply all local coefficients and their sharp vanishing range.
Three unequal integer frequencies	Proved continuation	Theorem 6.2 applies to disjoint singular grids. Its finite reduction includes all prescribed coordinate terms.

Result	Status here	Precise scope
Weighted colored kernels and Gamma correlations	Proved consequences	Finite root filtering and (7.7) give explicit identities; no minimal-basis or independence claim follows.
Current Gaussian S_6 and S_8 evaluations	Remain conjectural	No exact functional certificate for either short evaluation is provided here. Their inherited numerical proximity is not equality.

The first three source questions identified in Section 1 are answered under explicit hypotheses. The broader collision problem, minimal arithmetic bases for spectral derivatives, and the Gaussian evaluation conjectures remain separate questions. The current manuscript already treats S_4 as proved; it must not be reset to conjectural status when material from older arrivals is integrated [1].

8.2 A verified editorial correction already proposed upstream

The canonical integration chapter contains the wording *Basis atoms must be $\mathbb{Q}$ -independent* in its discussion of PSLQ [1, chapters/07-integration.tex]. The incoming Gauss–Hurwitz report already objects to that wording [2, Section 8]. That correction should be retained, with credit to the earlier report. An appropriate replacement is:

Known rational dependencies among the supplied constants should be removed or modeled explicitly. Otherwise PSLQ may return a pre-existing relation with zero target coefficient instead of an evaluation of the target.

This expresses the practical requirement without imposing a prior proof of rational independence on a discovery algorithm. A relation among the supplied atoms is a valid output of an integer-relation procedure [15]; it simply may not evaluate the additional target. The change is editorial and logical, not a new counterexample to a theorem.

8.3 Normalization points exposed by the new proofs

No new false theorem was found in the particular incoming results used as premises. The following are precise implementation and integration requirements established by this article.

1. **Differentiate the deformation before deleting zeros.** Equation (3.27) gives an exact diagnostic: $r_0(-1) = 0$ but its first spectral coefficient is $\pi^2/8$. The regularized Gamma product or the finite polynomial shift must be formed before taking a resonant derivative.
2. **Specify the sum and its reference coordinate.** The subtraction in (2.8) uses $n + a/2$. Replacing this reference without the finite transport (2.21) changes the counterterms. In the correlation problem, the prescribed ambient finite part gives the nonzero scalar term in (6.3); it cannot be replaced by a zero-mean pullback without its coordinate correction.
3. **Distinguish a pointwise derivative from a distributional derivative.** The nonlinear coefficient for $T_{n,r} = \text{FP } \gamma_n^{(r)}$ is $\alpha_{n,r}$, whereas that for $D^r G_n$ is $b_{n,r}$. Their difference is the exact contact term (5.20). The derivative comparison uses the common smooth completion of $\gamma_n(1+x)$ across the endpoint. Extending that entire smooth remainder by zero would introduce additional step-function derivatives and define a different object.

4. **Keep the transformed germ in composition.** The last term of (5.8) is $\mathfrak{A}_f[h \circ g]$. Leaving h unchanged there would discard the logarithmic and principal-part coefficients generated by the first coordinate change.
5. **Separate a negative-order value from its transverse derivative.** Equation (7.3) may be differentiated in A and B . A derivative in C needs additional information, supplied for the ordinary kernel by (7.4).
6. **Do not silently merge singular grids.** The full unequal-frequency theorem assumes disjoint singular sets. At collisions, products have higher pole order and the spectral integral has a different convergence domain. The coordinate law for a specified coincident product in Corollary 5.8 does not by itself identify the collision limit of separated finite parts.

These requirements do not identify hypothetical source errors. They state which objects the present proofs evaluate and which tempting changes would evaluate something else.

8.4 Finite verification and its evidentiary role

The supplied Python programs use exact SymPy algebra and mpmath arithmetic. The recorded runs use SymPy 1.14.0 and mpmath 1.3.0. Their machine-readable results are supplied with the sources.

Dougall and harmonic branch. `code/verify_dougall.py` checks the coefficient differential law and residue polynomial exactly through coefficient index two, as well as the explicit quartic C_1 . Its 19 numerical comparisons include generic resonances through $N = 3$, first and second resonant jets, a zero-residue case, the revived finite head, quartic moments, squared-binomial moments and harmonic sums, and (2.18). The run uses 75 decimal digits, the first 120 summands, and a finite centered asymptotic tail through C_{11} . The largest observed absolute discrepancy is 3.76169×10^{-36} , below the configured 10^{-31} threshold. The asymptotic-tail calculation is a diagnostic approximation; it does not furnish a rigorous remainder enclosure.

A separate unaccelerated check of (3.28) gives partial-sum errors of about 9.41×10^{-6} at 100 terms and 2.03×10^{-10} at 30,000 terms against the right side

$$-0.0147807743355189137009913877658542374\dots$$

This checks the summand directly without reusing the tail completion. The exact proof remains Theorem 3.1.

Nonlinear coordinate branch. `code/verify_nonlinear.py` records 138 exact finite-algebra checks of the coefficient generator, cocycle polynomial, low-order formulas, and filtration. The independent implementation `code/independent_cutoff.py` constructs the inverse coordinate series, transforms the test function with its inverse Jacobian, integrates each singular monomial in the source coordinate, and extracts the constant after composing the primitive. Its actual-cutoff side does not use the proposed residue formula or the $\alpha_{n,r}$ generator.

For

$$f(t) = q \left(t + \frac{2}{3}t^2 - \frac{1}{5}t^3 + \frac{1}{7}t^4 \right), \quad q = 1, 2,$$

it checks 48 pairs of Stieltjes and derivative indices, through $r = 3$ and $n = 2r + 2$, against 140 monomial test pairings. The $q = 2$ comparisons retain exact polynomial expressions in $\log 2$.

Every pairing also agrees with the reusable coordinate-coefficient implementation. These tests independently verify signs, Jacobians, and scale terms in low degrees.

Unequal-frequency branch. `code/verify_transport.py` records four exact polynomial integrals and 13 numerical comparisons at 45 decimal digits. The numerical checks include pointwise Stieltjes multiplication, the nonzero ambient means, direct finite-part quadrature at $(p_1, p_2, p_3) = (1, 2, 3)$, permutation symmetry, the simplified twelfth-root lower term, and the six-region Fourier normalization. For the latter, at orders $(2, 2, 2)$ the Hurwitz factors are Bernoulli polynomials. Direct piecewise polynomial integration for the recorded shifts and frequencies gives exactly

$$-\frac{1879}{125411328}.$$

A separate Mellin evaluation of the colored kernels agrees with it. The largest recorded scaled numerical discrepancy is about 1.46×10^{-45} .

A numerical differentiation issue. During verification, generic numerical differencing of $\zeta(s, 1/4)$ at $s = 0$ for the second order derivative lost accuracy in the selected backend. Native Hurwitz spectral derivatives, corroborated by Cauchy-circle differentiation, agreed with the independent harmonic sum. The supplied program therefore uses the native spectral derivative interface. This is a computational observation from the recorded run, not a correction to the analytic identity or a claim about every implementation.

The exact all-index results are proved by finite asymptotic remainders, normal convergence, analytic coefficient extraction, residues, and finite root filtering. Finite symbolic tests and floating-point agreement provide separate implementation evidence. No numerical file in this package is labeled an interval certificate.

8.5 Proposed repository integration

The modular TeX sources use the label prefixes `dg:`, `dp:`, `nc:`, `tr:`, and `tc:`. The source manifest pins the inspected baseline and records the incoming archive hashes. The companion `INTEGRATION_NOTES.md` gives a source-to-result crosswalk and proposed insertion points.

A suitable integration order is: retain the incoming Gauss theorem, append the centered Dougall extension and harmonic jets; place the primitive beside the existing Lerch and integration material; extend the affine finite-part section with the nonlinear law; and append the three-frequency transport after the existing pair and translated triple formulas. Each insertion should preserve its explicit subtraction convention and proof status. The source reports remain provenance, while the shared lemmas need appear only once in the consolidated manuscript.

9 Further research questions and concrete next targets

The proved results make the next questions more specific. The following agenda concerns exact identities, structural reductions, and proof certificates; it does not depend on searching for sharper inequalities. Questions are labeled as questions, rather than as unproved theorems.

9.1 Additional identities within the Dougall family

1. Closed polynomial formulas for all resonant counterterms. The recurrence (2.6) is already finite and exact, and (3.18) builds every quartic moment from one base sequence.

Can its coefficients be written in a useful closed combinatorial form in N, j , perhaps through central factorial numbers and Bernoulli polynomials? A useful outcome would provide both the subtraction polynomial and a compact exact certificate, avoiding large expanded rational polynomials. This is a representation question; the existence and the resonance value are proved here.

2. A minimal arithmetic presentation for rational-parameter jets. At rational a, b, c , Theorem 3.1 gives a finite list of Hurwitz derivatives and Stieltjes constants. Finite Fourier transforms then express rational-shift Hurwitz values in residue-class or Dirichlet-character coordinates. Which symmetries of a particular Dougall family eliminate entire character sectors? The first concrete case is the quartic family at shift $1/4$, comparing the Stieltjes constants $\gamma_m(1/4)$ with the zero-order spectral jets $\zeta^{(m)}(0, 1/4)$ from its half-resonance. An exact reduction matrix for each prescribed finite jet order would be useful. Numerical independence is a separate question.

3. Mixed parameter jets along residue-zero loci. Formula (3.1) makes all zeros explicit, but the article mainly illustrates derivatives in u . When b or c is a positive integer at most N , what short identities arise from mixed derivatives transverse and tangent to $\rho_N = 0$? The first target is a two-parameter table at $b = c = 1$, where the zero has higher multiplicity. The finite product predicts which derivative orders first survive. A complete statement should retain the small-index summands before specialization, just as in (3.27).

4. Primitive endpoint values beyond the counterterms. The weighted primitive in Theorem 4.1 lowers the hypergeometric order and gives a finite colored-polylogarithm description of the subtractions. It does not evaluate every endpoint of the remaining ${}_4F_3$. Which parameter choices admit an additional summation, a modular parametrization, or a finite iterated-integral reduction? The squared-binomial branch is a natural first case because its base generating function has classical integral representations. Any proposed polylogarithmic reduction should specify its function class and branches, rather than infer it from the appearance of polylogarithmic counterterms.

5. Other exactly summable well-poised families. The present proof uses two special structures: reflected Gamma shifts giving even powers, and an exact Gamma-product endpoint. Which further very-well-poised summations retain both structures under a one-parameter excess deformation? A precise first task is to identify a higher-order family, derive its centered finite asymptotic coefficients, and determine whether a differential transport identity simplifies every resonant constant as in (2.15). The existence of the Dougall result does not establish such a theorem for arbitrary hypergeometric families.

9.2 Coordinate algebra and collisions

6. Collision limits versus specified coincident finite parts. Let two translated singular points approach each other. Corollary 5.8 evaluates the coordinate change of the finite part of the pointwise product at coincidence, while Theorem 6.2 treats disjoint supports. What explicit local terms must be subtracted from the separated family to obtain that coincident finite part? The first nontrivial target is $\gamma_0(x)\gamma_0(x + \varepsilon)$, followed by $\gamma_m^{(r)}(x)\gamma_n^{(s)}(x + \varepsilon)$. The regulator and the relative approach direction must be fixed before taking the limit. Equality of two unspecified regularizations would not be a meaningful claim.

7. An intrinsic classification of the coordinate corrections. The residue law (5.8) is explicit. Can all local corrections compatible with scalar pullback, finite principal parts, and composition be classified, with the freedom of adding fixed delta derivatives made precise? The sharp filtration (5.32) provides a concrete family of graded invariants for that classification. A first step is to classify the finite jet action through pole order three and determine which coefficients can be changed by a fixed normalization.

8. Smooth coordinates and noninteger logarithmic exponents. For fixed pole order, the proof only uses finite coordinate jets and an integrable remainder. This suggests an extension from analytic coordinates to a stated finite differentiability class. The exact minimum regularity and remainder hypotheses should be proved, not inferred from formal Taylor manipulations. A distinct question is the extension of $\log^n x$ to noninteger powers of $\log x$, where branch choices and the polynomial filtration change. It should be treated separately from the integer Stieltjes hierarchy.

9.3 Colored kernels and exact relation certificates

9. Minimal finite reductions for three or more frequencies. The finite lift has at most $p_1 p_2 p_3$ translated triples, and each triple has six sign regions. Root filtering establishes existence of a finite ordinary-kernel presentation. Can common divisors, translation symmetry, and color conjugation compress this presentation canonically? The first target is an exact reduction algorithm for the $(1, 2, 3)$ example, with the order-derivative coordinates retained. For four frequencies, one should specify the resulting triple-sum kernels and a finite sign-region decomposition before claiming analogous closure.

10. Transverse derivatives at negative Tornheim order. At $C = -N$, the kernel reduces to products of ordinary polylogarithms. Equation (7.4) isolates the extra information in the first normal derivative. At roots of unity, which choices of A, B, N reduce that integral to a finite combination of known depth-two polylogarithms, order derivatives, and Gamma or Stieltjes data? The case $C = 0$ with A, B equal to zero or one is a concrete starting point. The ordinary negative-order value alone contains insufficient information.

11. Exact Gaussian S_6 and S_8 certificates. The retained conjectures need an exact relation in a specified functional or iterated-integral system, together with endpoint control and evaluation. The appropriate first task is to pin one candidate vector, its alphabet, and the exact relation generators. Either produce a certificate whose evaluation is the desired identity, or prove exclusion from that *specified* generated relation space. Exclusion from a selected formal space would identify a missing mechanism; it would not prove numerical independence or disprove the period identity. The current source ledger already emphasizes this distinction [1].

12. A common exact coefficient and certificate format. The Bernoulli recurrence, the nonlinear residue, and the finite root filter are all constructive operations on finite data once the analytic theorem has been fixed. A useful ProveIt implementation would record the theorem identifier, parameter domain, local coordinate, coefficient order, finite-head terms, and exact polynomial certificate in one reusable format. A first deliverable could replay (3.28), (5.31), and (7.12) using independent exact arithmetic. Such a format would support integration and auditing without treating a floating-point comparison as a proof certificate.

9.4 Research priorities

The shortest paths to additional exact identities appear to be mixed Dougall jets at explicit residue zeros, arithmetic reduction of the rational-shift special values, and normal derivatives of colored Tornheim kernels at $C = 0$. Collision limits require a separate normalization theorem. The Gaussian conjectures remain the most direct targets for a finite exact relation certificate, but the present continuation supplies no reason to mark them proved before that certificate exists.

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